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Paleomagnetic evidence for a Pleistocene clockwise rotation of the Calabro-Peloritan block (southern Italy)

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Abstract

A paleomagnetic study of marine marls of (Early) Pliocene age and clays of Late Pliocene to middle Pleistocene age from sites along the Ionian coast (Spartivento basin and Catanzaro depression of Calabria) and from sites along the Tyrrhenian coast of both Calabria and Sicily was carried out to determine the amount and timing of tectonic rotation of the underlying Calabro-Peloritan block.

The magnetisations of the Lower Pliocene Trubi marls have maximum blocking temperatures indicating the presence of magnetite. These Lower Pliocene sediments as well as the younger Upper Pliocene–Lower Pleistocene clays of this area reveal consistent clockwise rotations of 10–20°. There is evidence for a uniform 15° clockwise rotation of the entire Calabro-Peloritan block during the middle Pleistocene. This is opposite in sense of rotation to the earlier established 23° counterclockwise rotations for the Southern Apennines, but the timing is the same: between 0.7 and 0.2 Ma. The tectonic rotations occur 1 Myr after the major phases of extension in the southeastern part of the Tyrrhenian basin, but simultaneously with the onset of major uplift of the block.

1. Introduction

Magnetostratigraphic studies of Pliocene and Pleistocene marine marls/clays of southern Italy have revealed the existence of stable and consistent paleomagnetic directions. These paleomagnetic results have been used to establish a reliable magnetostratigraphy, with an emphasis on dating biostratigraphic datum levels and paleoclimatic changes in the Mediterranean area (Hilgen and Langereis, 1988; Zachariasse et al., 1989, 1990; Langereis and Hilgen, 1991; Zijderveld et al., 1991) and they have formed the basis for the astronomically calibrated polarity time scale for the last 5.5 Myr (Hilgen, 1991a,b). In the present study, the consistent directions are used to determine tectonic rotations of the different Plio-Pleistocene formations along the Calabrian arc.

In the Central Mediterranean area, this Calabrian arc links the Maghrebides of North Africa and Sicily to the Italian Apennines. The Tyrrhenian basin is at the internal side of the Calabrian arc. Opening (subsidence) of this basin started in the Tortonian and is continuing at the present day in the southeastern part of the Tyrrhenian basin and on-shore Calabria (Kastens et al., 1988; Knott and Turco, 1991). Two major phases of extension caused formation of oceanic crust: (1) during the latest Miocene in the central Tyrrhenian (Vavilov sector); and (2) during the Early

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Pleistocene (2.0–1.6 Ma) further to the southeast (Marsili sector) (Sartori, 1990).

The central part of the Calabrian arc, the so-called Calabro-Peloritan block, moved in southeastern direction from the early Miocene onward (Malinverno and Ryan, 1986). This block originates from the southern European margin of the Tethys basin (Alvarez et al., 1974) and can be considered as a ("boat-shaped") element that overrides the Ionian foreland (Van Dijk and Okkes, 1991). Roll-back of a sinking slab across the Western Mediterranean detached the Calabro-Peloritan block from the South European margin (Malinverno and Ryan, 1986). This frag-

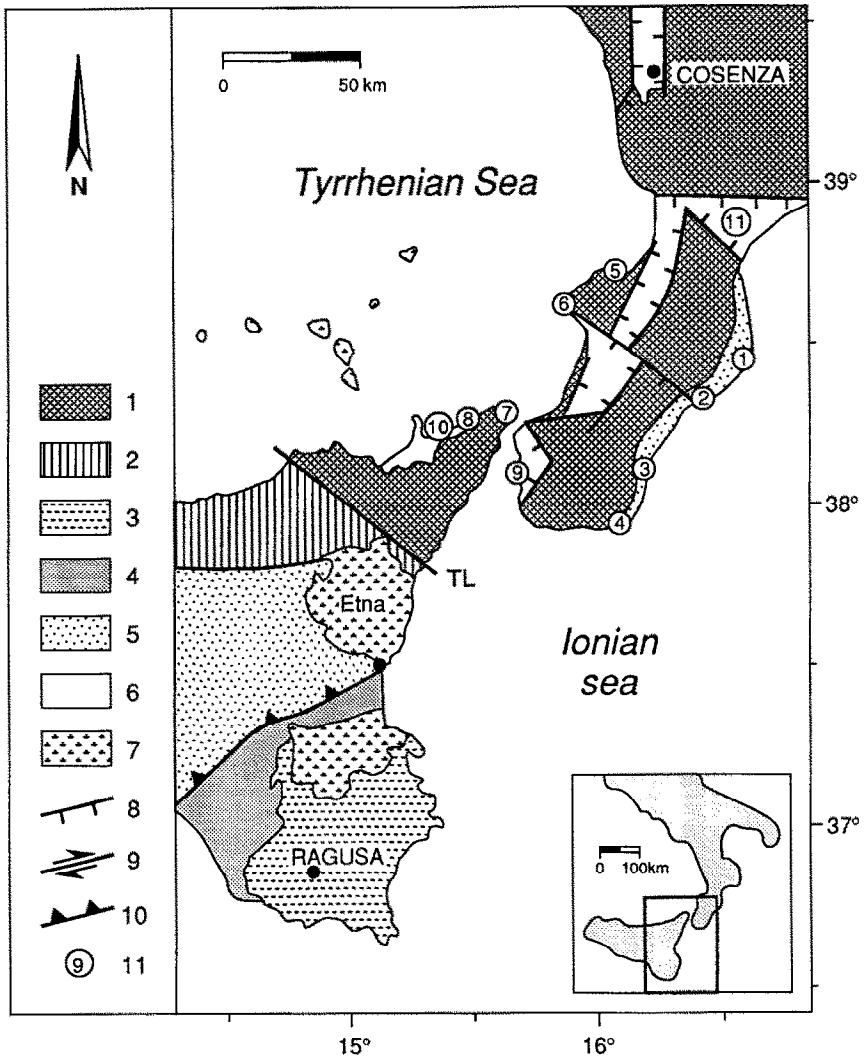


Fig. 1. Structural map of the southern part of the Calabro-Sicilian arc with the distribution of the localities sampled in this study. *1* = Calabro-Peloritan units; *2* = northern Sicilian belt (Maghrebides); *3* = Ragusa platform (foreland); *4* = Gela–Catania foredeep; *5* = external Plio-Pleistocene basins; *6* = internal Plio-Pleistocene basins; *7* = volcanics; *8* = normal fault; *9* = fault; *10* = deformation fault; and *11* = sampled locality. *TL* = Taormina line. Localities (with abbreviations): *(1)* = Singa section (CS); *(2)* = Monte Limina ML); *(3)* = Careri section (CAR); *(4)* = Capo Spartivento (CP); *(5)* = Paradísoni (PD); *(6)* = Capo Vaticano (VAT); *(7)* = Messina (MES); *(8)* = Villafranca (VIF); *(9)* = Croce Valanidi (CV); *(10)* = Milazzo (MIL); *(11)* = Catanzaro area: Caraffa di Catanzaro (CC) and Santa Maria di Catanzaro (SMC).

mentation of the South European margin of the Tethys basin starting in the Early Miocene (Cohen, 1980) is paleomagnetically evidenced by a counterclockwise rotation of Sardinia (Montigny et al., 1981). Our study investigates possible tectonic rotations and their timing for the Calabro-Peloritan block during the final stage of this process. It has been suggested recently that tectonic rotations in the Calabrian arc are caused by pinning of the moving Calabro-Peloritan block between two continental promontories, the Adria block to the northeast and the Ragusa platform to the southwest (Malinverno and Ryan, 1986; Knott and Turco, 1991). In this model, tectonic rotations are coeval with the major phases of extension.

Earlier paleomagnetic studies in Neogene rocks of the Calabrian arc suggest for the Messina strait area a total amount of 36° clockwise rotation relative to the magnetic north since the Early Pliocene (Aïfa et al., 1988). In the external foredeep basin of Sicily (Caltanissetta basin), Lower Pliocene marls reveal a 34° clockwise rotation relative to the magnetic north (Scheepers and Langereis, 1993); 19–24° of this rotation is younger than 2.8 Ma, and 5–10° took place around 2.9 Ma. For the present paper, Lower Pliocene marls of the Trubi formation in the Spartivento basin of southern Calabria and along the Tyrrhenian coast of both Calabria and Sicily were studied to investigate the total rotation since the Early Pliocene (Fig. 1). In addition, the results from younger (Pleistocene) clays throughout the area are discussed.

2. Sampling and methods

For most outcrops, removal of the weathered surface was necessary to reach fresh material. Per site, we collected four to seven cylindrical standard cores using an electric drill and portable generator.

In the laboratory, routine methods were applied in order to obtain characteristic remanent magnetisation (ChRM) directions for the clays. The natural remanent magnetisations (NRM) of at least one sample per core was analysed by means of progressive stepwise thermal demagnetisation using small temperature increments (30–50°C). The NRM was measured both on cryogenic magnetometers (ScT and 2G) and on a spinner magnetometer (JR-3) with a digital interface.

Demagnetisation diagrams (Zijderveld, 1967) were used to analyse the directional stability for each demagnetised sample. Several methods have been applied to determine a reliable ChRM direction. The *direct method* means that magnetisation components were determined by least square fitting of lines through at least four data points (Kirschvink, 1980). For some samples, the NRM does not linearly decrease to the origin, implying that two NRM components with overlapping blocking temperature spectra are present. In that case great circles were determined. The *great circle method* uses the great circles of one site to calculate the "hidden" component (McFadden and McElhinny, 1988). A combination of direct and great circle methods ("*combination method*") is used if (only) some samples from the same sites have a well-determined direction ("setpoint" according to McFadden and McElhinny, 1988). Finally, the *stacking method* (Scheepers and Zijderveld, 1992) is used when sites show a scatter behaviour because of low intensities and when there is not a clear secondary magnetisation.

3. The (Lower) Pliocene marls

The Lower Pliocene marls from the Trubi formation covering the Calabro-Peloritan block are sediments which carry a stable primary magnetisation (Zijderveld et al., 1986). For the Caltanissetta basin of Sicily the Lower Pliocene marls show on average a 34° clockwise rotation (Scheepers and Langereis, 1993). The same formation was investigated along the external (Ionian) part of the Calabro-Peloritan block and along the internal (Tyrrhenian) part of the same block in both southern Calabria and Sicily.

3.1. The Ionian coast

In the Spartivento basin along the Ionian coast of southern Calabria (Fig. 1), the marls of the

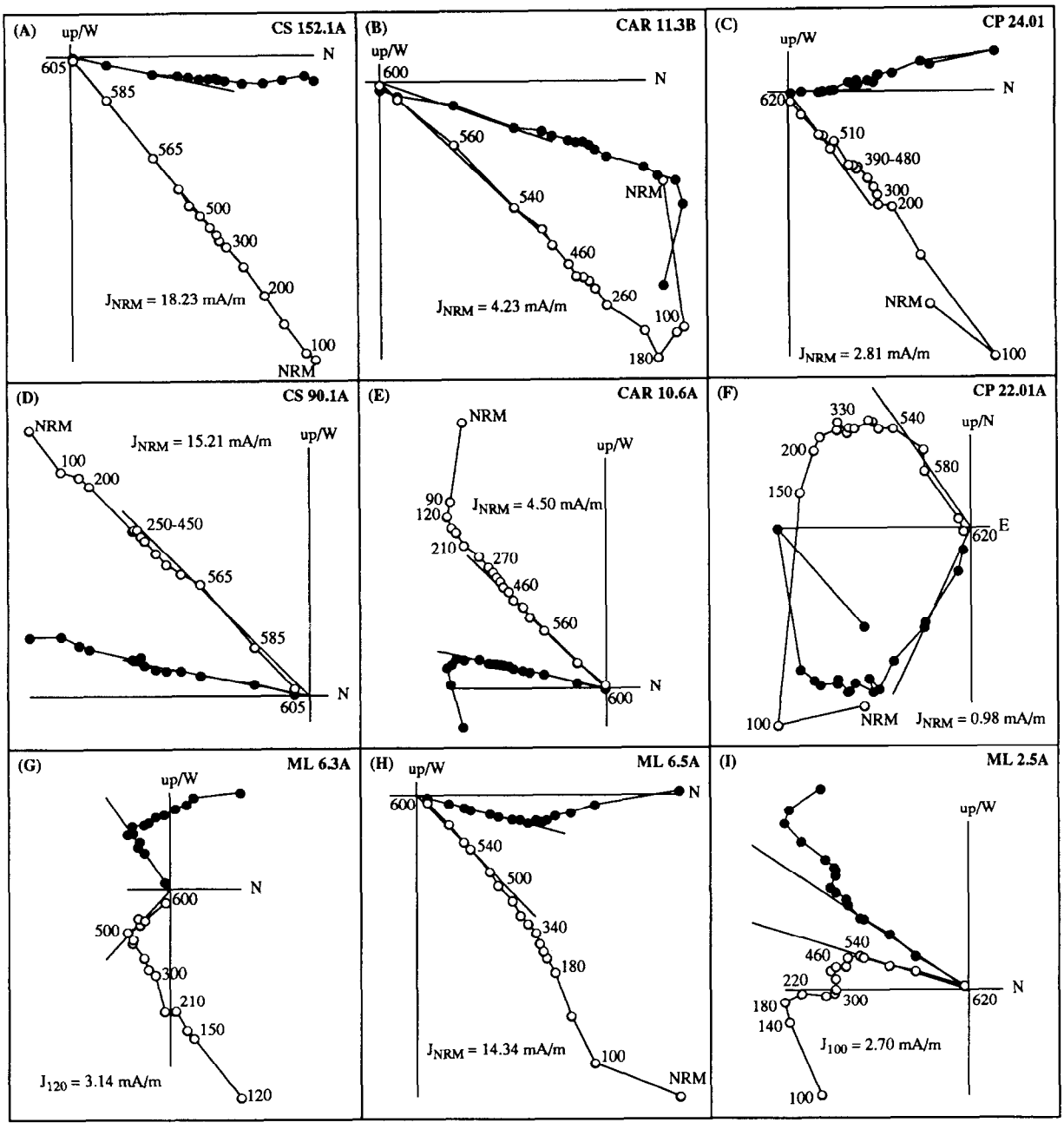


Fig. 2. Representative thermal demagnetisation diagrams of marls from the Trubi formation (Ionian coast southern Calabria, Spartivento basin). ● represent vector end-points projected on the horizontal plane (declination); ○ are vertical plane projections (inclination). *CS* = Singa; *CAR* = Careri; *CP* = Capo Spartivento; *ML* = Monte Límina. Numbers in diagrams denote demagnetisation steps in °C. Direction and intensity of magnetisation during thermal demagnetisation were analysed by vector diagram representation (Zijderveld, 1967); lines represent best-fitting lines through data points using principal component analysis (Kirschvink, 1980).

Trubi and Narbone formations are exposed. The basin belongs to the external part of the Tyrrhenian arc and is covering the Calabro-Peloritan block. Tectonically it is considered as a ridge of imbricated Pliocene and Pleistocene sedimentary rocks dissected by NW–SE and NE–SW-trending lineaments into a number of segments (Van Dijk and Okkes, 1991). The ridge is considered to be a submarine accretionary wedge (Kastens, 1981) and links the Apennines to the north and the Maghrebides to the west. These marls were sampled at four different localities in the Spartivento basin: (1) the Monte Singa section; (2) Monte Límina; (3) the Careri section; and (4) the Capo Spartivento section (Fig. 1). The fresh sediments of all localities are blue-grey coloured.

The Monte Singa area is situated 3 km northwest of Riace Marina in the northern part of the Spartivento basin (Fig. 1). The section covers a considerable part of the Pliocene and includes the Miocene–Pliocene boundary as well as the Pliocene–Pleistocene boundary, but in the middle of the Pliocene, a 1 Myr hiatus occurs (Zijderveld et al., 1986, 1991). The lower Singa section represents the Early Pliocene Trubi formation and was deposited between 4.9 and 4.0 Ma, while the upper Singa section represents the Narbone formation, deposited between 2.5 and 1.9 Ma. Bedding tilt is approximately 5° to the south-southeast.

The outcrops at Monte Límina and the Careri section are in the central part of the Spartivento basin (Fig. 1). Monte Límina is almost 900 m above sea level, approximately 5 km west-northwest of Mámmola. The Trubi marls unconformably overly continental conglomerates of Messinian age; bedding tilt is 17° to the south-southwest. The section was sampled at seven sites/levels. The marls have an Early Pliocene age (see Appendix) and according to the established magnetic polarities the age is between 4.9 and 4.6 Ma. The Careri section is further south, approximately 150 m above sea level, 5 km west of Bovalino Marina; bedding tilt is to the south-southeast. The Careri section was sampled at thirteen different levels. The sampling interval in the Careri section represents the lower part of the Trubi formation (see Appendix); the age is between 4.6 and 3.7 Ma. The Capo Spartivento section is located in the most southern part of the Spartivento basin; horizontally bedded Trubi marls overly non-marine coarse conglomerates of Messinian age. The section is in the Gilbert Chron, and contains the Thvera and Sidufjall subchrons; it ranges in age from 4.8 to 4.3 Ma (Channell et al., 1988; Hilgen and Langereis, 1993).

Demagnetisation results

For the Trubi marls in the Spartivento basin the NRM magnetisation is removed at 600–620°C (Figs. 2, 3); the component removed from 500 up to this 600–620°C is defined as a high-temperature (HT) component. There are clearly two types of demagnetisation behaviour from these diagrams. The samples from both the Singa and the Careri section (Fig. 2A, B, D, E) show that from 200–250°C onwards the demagnetisation diagrams have a linear decrease towards the origin. The samples from both Monte Límina and Capo Spartivento have a component removed between 100 and 500°C (Fig. 2F, I) or even higher (540°C). This component has a normal polarity and is probably due to weathering considering its present-day field direction. It is clearly a secondary component. In many cases, only the last few temperature steps show a linear decrease towards the origin. Apparently, there is a large blocking temperature overlap between the two components (see also Van Velzen and Zijderveld, 1990).

Sites showing no overprint

The lower Singa section. The lower part of the Singa section has negligible secondary components (Fig. 2A, D) and the HT component of the samples was determined by least square fitting of lines through all data points between 500 and 600°C (Fig. 2A, D). The individual samples show consistent clockwise deviations for both normal and reversed polarity (Fig. 3A, D). To give the best estimate for the rotation of the section a mean HT direction was calculated for every 5 m of the section (Table 1; Fig. 3B). If there is a change in polarity within this 5 m, a mean direction was calculated for a smaller interval (Table

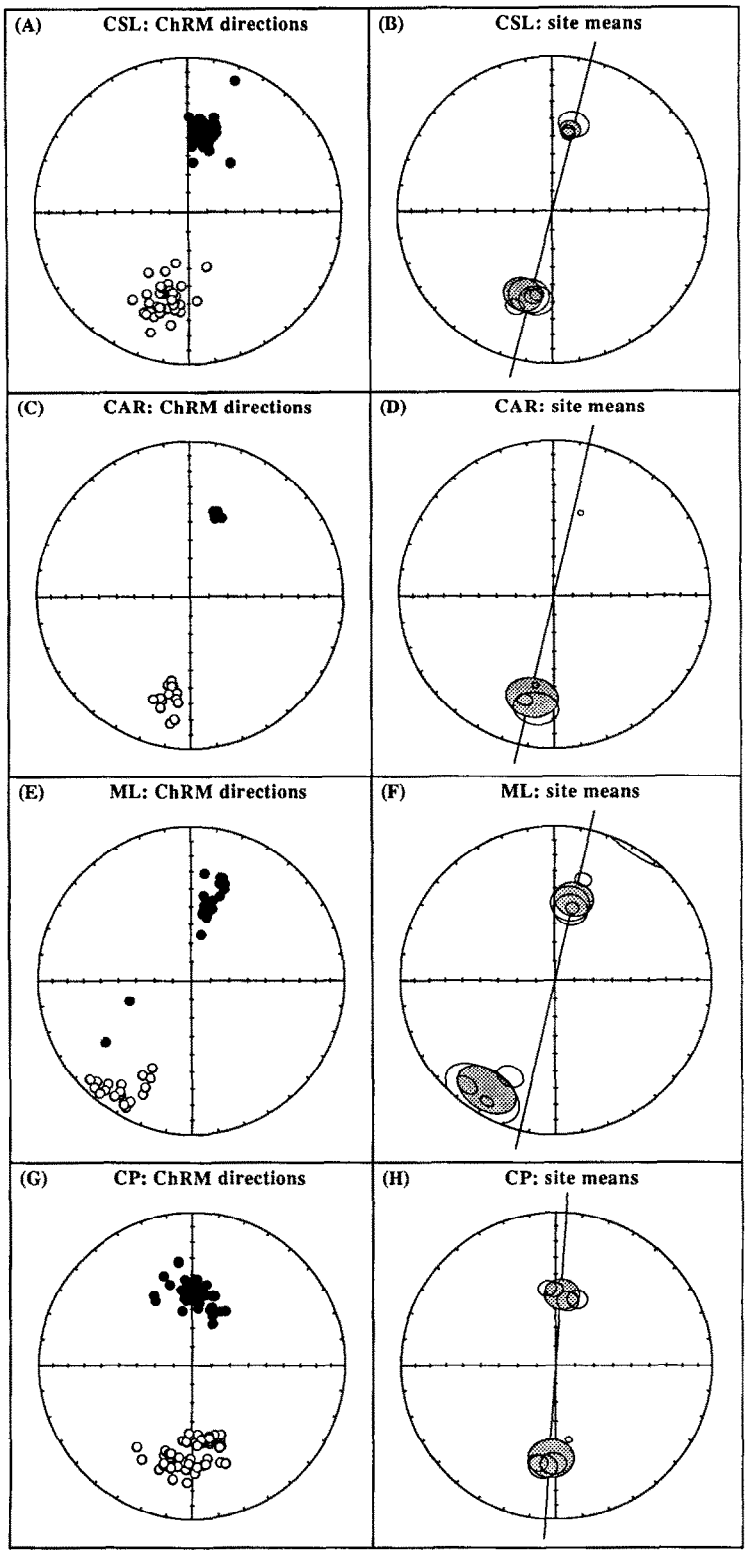

(A) CSL: ChRM directions
(B) CSL: site means
(C) CAR: ChRM directions
(D) CAR: site means
(E) ML: ChRM directions
(F) ML: site means
(G) CP: ChRM directions
(H) CP: site means

Table 1
Paleomagnetic results of the Singa and Capo Spartivento sections using Fisher (1953) statistics; a mean direction is calculated for stratigraphic intervals (lower Singa: every 5 m, upper Singa: every 20 m, Capo Spartivento: every 4 m); at a polarity break the interval is smaller

Level (m)	d (m)	Method	N	D(tc)	I(tc)	k	α_{95}
Lower Singa section							
0.8– 2.3	1.5	direct	7	201.3	−33.4	178	4.5
7.5– 11.8	4.3	direct	12	13.7	42.0	36	7.3
16.0– 17.0	1.0	direct	3	190.4	−40.1	221	8.3
22.6– 27.5	4.9	direct	10	190.5	−43.9	206	3.4
29.3– 32.3	3.0	direct	10	192.9	−42.3	141	4.1
35.0– 39.8	4.7	direct	14	11.2	46.2	175	3.0
40.0– 42.5	2.4	direct	15	11.5	48.1	126	3.4
44.7– 48.9	4.3	direct	11	195.6	−42.1	23	9.8
N intervals			3	12.2	45.4	615	5.0
R intervals			5	195.6	−40.5	139	6.5
All intervals			8	14.4	42.4	173	4.2
Upper Singa section							
All R1 diagrams		direct	36	191.7	−50.0	28	4.6
0.0– 25.0	25.0	direct	8	352.7	46.3	126	5.0
25.0– 45.0	20.0	combined	13	186.2	−52.2	33	7.3
45.0– 65.0	20.0	combined	10	191.8	−51.1	62	6.2
65.0– 85.0	20.0	combined	11	206.8	−53.7	20	10.6
85.0–105.0	20.0	combined	15	190.6	−46.6	25	8.2
R intervals			4	193.6	−51.1	166	7.2
All intervals			5	9.1	50.5	91	8.1
Capo Spartivento section							
All R1 diagrams		direct	22	192.1	−34.6	89	3.3
All N1 diagrams		direct	31	3.7	50.7	55	3.5
0.0– 3.3	3.3	great circle	33	170.4	−48.9	101	2.5
3.3– 7.3	4.0	direct	10	10.7	55.0	79	5.5
7.3– 11.3	4.0	direct	7	18.2	52.7	143	5.1
11.3– 14.5	3.2	direct	18	353.9	48.0	61	4.5
14.5– 18.5	4.0	combined	7	181.5	−36.0	58	8.0
18.5– 20.2	1.7	combined	19	190.5	−35.6	99	5.2
20.2– 23.5	3.3	direct	9	359.0	48.9	141	4.4
23.5– 24.6	1.1	direct	6	187.4	−34.0	100	6.7
N intervals			4	4.9	51.5	112	8.7
R intervals			4	183.1	−38.8	72	10.9
All intervals			8	3.9	45.2	59	7.2

Level gives the stratigraphic position (begin and end) for the interval; d = length of interval; method = method of calculation of mean direction (see text); N = number of samples used for calculation of mean direction; D(tc), I(tc) = declination, inclination, respectively, after bedding tilt correction; k = precision parameter; α_{95} = cone of confidence at the 95% level.

Fig. 3. Equal-area projection of HT component of samples from the Lower Pliocene marls of the Spartivento basin along the Ionian coast (after bedding tilt correction). In the diagrams to the left the circles represent the individual directions; • (○) denote downward (upward) projections, respectively. Part of the Capo Spartivento directions was determined with the method of McFadden and McElhinny (1988). In the diagrams to the right the circles give Fisher's (1953) α_{95} for the different site means. The shaded area gives the α_{95} of the overall mean direction using the site/interval mean directions. The lines in the right diagrams represent the estimated tectonic rotation of the outcrop.

Table 2
Paleomagnetic results from the sites in Trubi marls covering the Calabro-Peloritan block using Fisher (1953) statistics

Site	Method	N	D(tc)	I(tc)	k	α_{95}	Int
Careri section (Spartivento basin)							
CAR 10	direct	8	191.5	−39.9	1133	1.6	2.76
CAR 11	direct	8	18.1	43.6	1586	1.4	3.45
CAR 12	direct	6	195.5	−29.8	346	3.6	0.62
CAR 13	direct	4	189.0	−25.5	83	10.2	6.89
Sites (R)		3	192.0	−31.8	105	12.1	
Sites (NR)		4	13.3	34.8	80	10.4	
Monte Límina (Spartivento basin)							
ML 1	direct	5	209.3	−10.2	705	2.9	2.39
ML 2	direct	7	219.8	−11.5	161	4.8	1.86
ML 3	direct	6	11.8	49.1	70	8.1	1.94
ML 4	direct	6	15.8	32.8	300	3.9	6.48
ML 5	direct	3	11.3	48.0	165	9.6	2.05
ML 6up	direct	6	13.7	50.1	465	3.1	7.22
ML 6mid[a]	direct	2	241.7	43.3	19	61.1	0.92
ML 6down	direct	3	212.7	−119	56	16.6	0.61
ML 7	direct	4	204.9	−31.2	222	6.2	0.63
Sites (N)		4	13.3	45.0	95	9.5	
Sites (R)		4	211.9	−16.2	49	13.3	
Sites (NR)		8	24.0	31.0	17	13.9	
Paradísoni (Tropea), quarry [A]							
PD 1[a]	great circle	4	173.6	−55.9	5	47.2	0.05
PD 2	great circle	6	192.7	−52.3	13	19.5	0.06
PD 3	great circle	6	187.1	−46.5	47	9.9	0.04
PD 4	great circle	6	190.5	−45.1	21	14.8	0.05
PD 1–4	great circle	22	185.2	−49.8	30	5.8	0.05
PD 5	direct	5	6.9	55.3	17	19.0	0.15
Sites (R)		3	190.0	−48.0	364	6.5	
Sites (NR)		4	9.3	49.8	250	5.8	
Paradísoni (Tropea), quarry [B]							
PD 7	direct	5	14.6	52.8	28	14.7	0.05
PD 8	great circle	7	185.6	−49.5	17	15.2	0.07
PD 9	great circle	7	186.2	−43.7	1424	1.6	0.27
Sites (NR)		3	8.6	48.7	207	8.6	
Paradisoni (Tropea)							
All sites (N)		2	10.9	54.1	493	11.3	
All sites (R)		5	188.3	−47.5	407	3.8	
All sites (NR)		7	9.0	49.4	272	3.7	
Capo Vaticano section (Tropea)							
VAT 2	direct	2	21.2	54.6	570	10.5	2.60
VAT 3	stacking	6	14.2	52.6	mad	5.1	O.O5
VAT 4	stacking	3	24.7	41.2	mad	5.5	0.16
VAT 5	stacking	6	23.1	50.6	mad	6.1	0.07
Sites (N)		4	21.0	49.8	149	7.5	
Lower Pliocene Tropea area (PD and VAT)							
Sites (N)		6	17.8	51.3	148	5.5	
Sites (R)		6	186.2	−49.0	170	5.2	
Sites (NR)		11	13.3	49.7	153	3.7	
Areas (PD/VAT)		2	15.0	49.8	217	17.0	

Table 2 (continued)

Site	Method	N	D(tc)	I(tc)	k	α_{95}	Int
Messina area (Sicily)							
MES 1–3	great circle	8	188.5	−49.6	11	17.2	0.04
MES 4	combined	6	188.0	−51.7	19	15.9	0.07
MES 5	great circle	7	194.4	−47.1	23	12.8	0.02
MES 1–5	combined	21	190.8	−50.5	23	6.8	0.04
Sites (R)		3	190.4	−49.5	610	5.0	
Villafranca area (Sicily)							
VIF 2–3	great circle	5	174.2	−49.8	66	9.5	0.03
VIF 4	direct	6	215.3	−51.0	139	5.7	3.18
VIF 5	stacking	7	236.4	−58.8	mad	9.0	0.10
Sites (R)		3	206.8	−56.0	18	29.7	
Lower Pliocene Sicily (MES and VIF)							
Sites (R)		6	197.9	−53.0	35	11.5	
Areas (MES/VIF)		2	18.0	53.0	94	26.0	

[a] Denotes (mean) direction not used to calculate overall mean direction for the locality. Other symbols are the same as in Table 1. Int = intensity for the calculated component (in mA/m).

1), because we want to separate between normal and reversed polarity mean directions (reversal test). There are three normal intervals in the section giving a mean direction of $D_m = 12.2°$ and $I_m = 45.4°$ ($k = 615$) (Fig. 3B) giving equal weight to the interval means. Further, there are six reversed intervals giving an overall mean direction of $D_m = 199.5°$ and $I_m = -41.8°$ ($k = 51$) (Fig. 3B). The angle between normal and reversed polarity mean directions is 5.5° here and the classification of the reversal test is "B" (McFadden and McElhinny, 1990). The mean direction of all nine intervals gives a 14° clockwise rotation for the lower part of the Singa section (Table 1).

The Careri section. From the Careri section, the upper nine sites (CAR 1–9) have very low (50–200 μA/m) initial NRM intensities; hardly any reliable HT directions could be determined. The polarities are clear (reversed), but the measurements show no decrease to the origin. The lower four sites of the section (CAR 10–13) give good results and similar demagnetisation behaviour as the lower Singa section (Fig. 2B, E). Three of these sites have a reversed polarity, one is of normal polarity (CAR 11). A small secondary magnetisation is removed at temperatures below 200°C (Fig. 2B, E).

The individual samples have clockwise declinations for both normal and reversed polarity (Fig. 3C) and the mean direction of all four sites of the lower part of the Careri section shows a clockwise rotation of 13° (Table 2; Fig. 3D). For the Careri section, the reversed polarity mean direction is $D_m = 192.0°$ and $I_m = -31.8°$ ($k = 105$). The angle between the normal polarity site and the reversed polarity mean direction is 12.8°; classification of the reversal test is "Indeterminate". The sites from both the Singa and the Careri section have mean inclinations which are deviating considerably from the inclination (57.5°) of the geocentric axial dipole (GAD) field for the present latitude of the locality, which may have been caused by early dewatering and compaction.

Sites showing a secondary overprint

The Monte Límina section. The samples from the Monte Límina section show maximum blocking temperatures of 580–600°C as well. In contrast to the samples from the Careri and Singa sections, there is a large secondary magnetisation (e.g., Fig. 2I). The HT direction of some sites is probably affected by a large overlap (up to 500–540°C) with the blocking temperatures of the secondary component, as can be seen from the low inclina-

tion values for some of the HT magnetisations (Table 2). The lowest site of the section (ML 6) includes cores with different polarities of the HT component; the lower part of this site is reversed, the upper part normal (Fig. 2H), whereas the middle part shows intermediate directions (Fig. 2G), but all show the same secondary magnetisation. According to the magnetobiostratigraphic constraints, this site was taken in the lower Thvera polarity transition. The next three sites of the section (ML 7 and ML 1–2) are reversed with very shallow inclination values (10–30°, Table 2; Fig. 3F), while the upper three sites of the section have a normal polarity HT component. The mean direction of the normal polarity sites gives D_m = 13.3° and I_m = 45.0° (k = 95), similar to the observations from the Singa and Careri sections. The mean direction of the four reversed polarity sites results in a mean direction with D_m = 211.9° and I_m = −16.2° (k = 49). It is clear that we have HT directions that are far from anti-parallel; the angle between the two groups is 32.8°, giving a negative reversal test. Further, the reversed polarity sites have anomalously low inclination values, probably related to a complex interference of NRM components (Fig. 2I); we did not use the reversed polarity sites as an estimate for the tectonic rotation. The normal polarity HT magnetisations (Fig. 3F) give a 13° clockwise rotation, in good accordance with the results from the

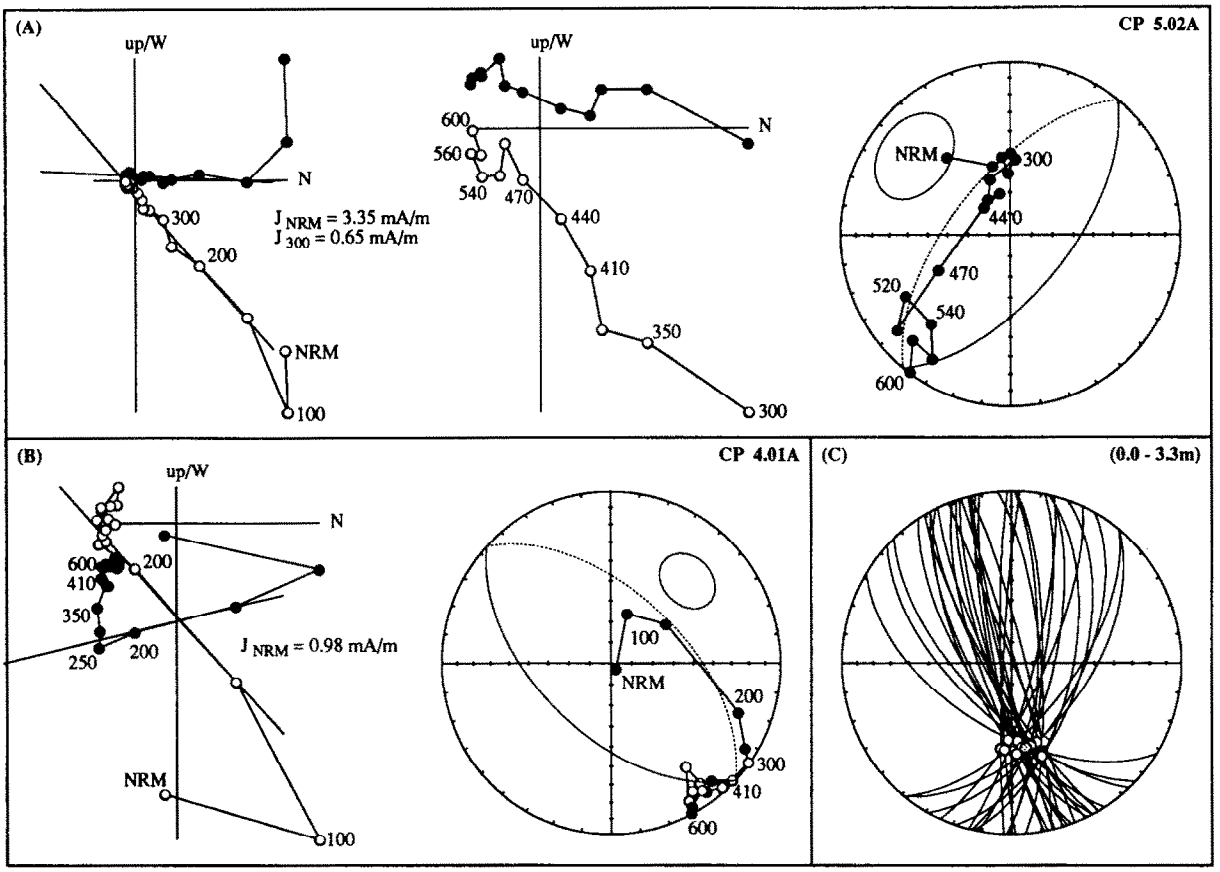


Fig. 4. Thermal demagnetisation diagrams of two samples from the Capo Spartivento section showing no final linear decay to the origin. All the great circle planes were calculated for the problematic samples; the great circle planes were combined to calculate a common direction for the lowest interval of the section (C). The ellipses in the equal-area projection show the maximum angular deviation of the calculated great circle planes. Other symbols as in Fig. 2.

Careri (13°) and Singa (14°) sections, which are south and north of Monte Límina (Fig. 1).

The Capo Spartivento section. Demagnetisation results from the Capo Spartivento section (Hilgen and Langereis, 1993) also show a large overlap between secondary and HT components (Fig. 2F). Our method to calculate a reliable mean direction for this section was complicated by the diversity in demagnetisation behaviour. Initially, we selected the demagnetisation diagrams—for both normal and reversed polarity—showing a clear linear decay to the origin and a minor secondary component. Calculation of the mean HT component shows that the mean direction from the 22 samples with reversed polarity (R1) and the mean direction from the 31 samples with normal polarity (N1) have an angle of 17.2°; there is a negative reversal test.

Many of the samples from this section, however, show demagnetisation diagrams that never reached a stable end vector (Fig. 4A, B), especially in the lower part (0.0–3.3 m) of the section. Therefore, we calculated the great circle planes through the data points for these samples (Fig. 4A, B). The planes with maximum angular deviations below 25° were used to calculate the hidden HT component for this interval (Fig. 4C). For the interval between 14.5 and 20.2 m, we used a combination of direct observations and great circles (McFadden and McElhinny, 1988). Calculation of mean directions (per 4 m) results in four normal polarity and four reversed polarity means (Table 1). With this procedure, the HT directions show a small clockwise deviation for both normal and reversed polarity. The mean direction of the four normal polarity intervals (not using great circles) gives $D_m = 4.9°$ and $I_m = 51.5°$ ($k = 112$); the mean direction for the four reversed polarity intervals (partly using great circle planes) gives $D_m = 183.1°$ and $I_m = -38.8°$ ($k = 20$). These mean directions make an angle of 12.8°, the reversal test is still negative. Because the declinations are nearly antipodal, however, we assume that the average declination represents a 4° clockwise rotation of this section (Table 1; see also Fig. 3H).

3.2. Trubi along the Tyrrhenian coast

To examine whether the clockwise rotation observed in the Pliocene marls of the external Ionian part of the Calabro-Peloritan block is also present along the Tyrrhenian coast, four localities in the Trubi formation along the Tyrrhenian coast were sampled in both southern Calabria and Sicily (Fig. 1). This coast area is—at the present day—characterised by extension (Knott and Turco, 1991). Along the Tyrrhenian coast of southern Calabria, the Trubi formation was sampled at two different localities on the Tropea peninsula: near (5) Paradísoni and at (6) Capo Vaticano (Fig. 1). Along the Tyrrhenian coast of Sicily we collected from two distinct areas: near (7) Messina and near (8) Villafranca (Fig. 1).

The Capo Vaticano section is 8 km south-southwest of Tropea (southern Calabria) directly along the Tyrrhenian coast. We sampled seven sites/levels in Lower Pliocene marls with a minimum (2°) tilt to the south; the age of the sampled interval is between 4.6 and 4.1 Ma (see Appendix). Five sites (VAT 1–5) were sampled in the lower part along the beach; two sites (VAT 6–7) in the upper part of the section. The other locality on the Tropea promontory, near Paradísoni, is approximately 10 km east-northeast of Tropea. Here, we sampled two different quarries in Lower Pliocene Trubi marls. One quarry is located west of the road from the village to the coast; in this quarry PD[A] we sampled six sites/levels (PD 1–6); bedding tilt is slightly to the northeast. Four sites were taken in the lower part of the quarry (PD 1–4) and two sites in the upper part of the quarry (PD 5–6). The age of the sampled interval is between 4.8 and 4.6 Ma (see Appendix). The other quarry near Paradísoni is east of the same road from the village to the coast; in this quarry PD[B] we sampled three sites/levels (PD 7–9); bedding tilt is 11° to the north. Along the Tyrrhenian coast of Sicily, we further collected four sites in the area between Villafranca and Rometta. Two sites just south of Villafranca (VIF 2–3) in marls of Lower Pliocene age dipping 9° to the north, and directly north of Rometta two sites (VIF 4–5) in blue-grey coloured

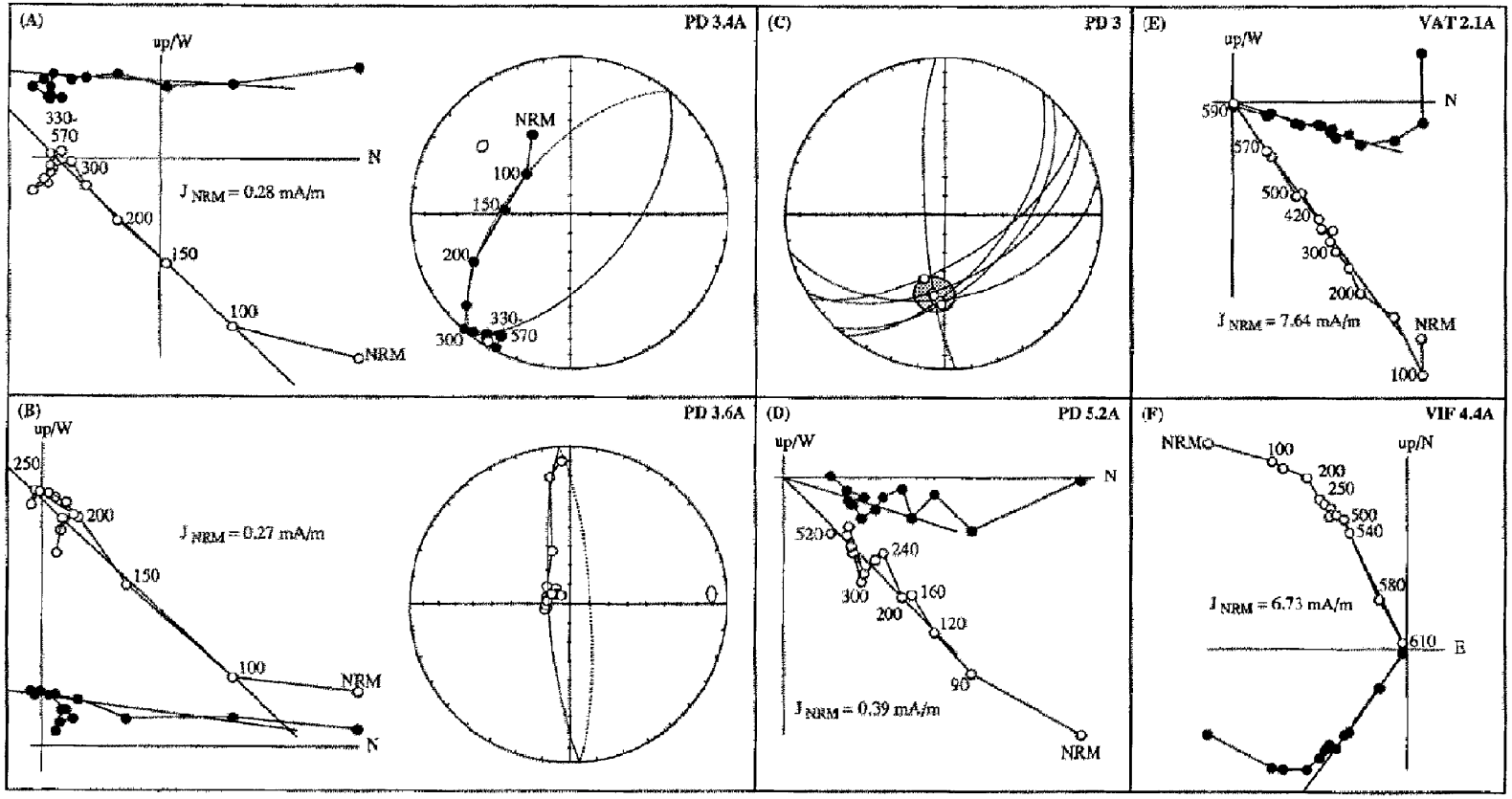


Fig. 5. Representative thermal demagnetisation diagrams of marls from the Trubi formation along the Tyrrhenian coast of both Calabria and Sicily. Symbols the same as used in Figs. 2 and 4. (C) gives the calculation of the mean direction of site *PD 3* using the great circle planes. *PD* = Paradisoni; *VAT* = Capo Vaticano; *VIF* = Villafranca.

silts of Late Pliocene age. Bedding is slightly (4–6°) tilting to the west. Further east, in the area north of Messina (Fig. 1), we sampled five sites (MES 1–5) in marine marls of Early Pliocene age from two different outcrops. Sites MES 1–3 are from one outcrop; bedding tilt is 12° to the north, whereas sites MES 4–5 are from another outcrop dipping 12° to the east.

Demagnetisation

From the 25 sites of four different localities along the Tyrrhenian coast only two sites give stable magnetisations having a linear decay to the origin (Fig. 5E, F); site VAT 2 in the Lower Pliocene Capo Vaticano section (southern Calabria) and site VIF 4 of Late Pliocene age in the Villafranca area (Sicily). Maximum blocking temperatures are the same as found in samples from the Trubi along the Ionian coast, between 590 and 610°C. The normal polarity site in the Capo Vaticano section shows a 21° clockwise rotation, whereas the reversed polarity site in the Villafranca area shows a 35° clockwise rotation (Table 2). Site PD 5 in the Paradísoni area shows a clear clockwise rotation of 7° (Table 2; Fig. 5D); some scatter produces a NRM decay that is not along a perfect line. For the other 22 sites, we applied alternative methods to derive the best estimate for the tectonic rotation.

The Paradísoni area. For the Paradísoni area, the quarry PD[A] containing sites PD 1–6 gives very poor paleomagnetic results; initial NRM intensities are very low (between 20 and 90 μA/m), even though the polarity is always very clear. Therefore, we use great circles to derive final magnetisation directions per site. For example, site PD 3 shows demagnetisation diagrams with a clearly reversed polarity final component (Fig. 5A, B). Because the diagrams show a cluster at the highest temperatures we cannot determine a reliable HT component. Great circle planes are all well determined: maximum angular deviations are below 10°. The trend of the magnetisation vector in the great circle planes is always in the same direction, although the orientation of the great circle planes varies considerably (Fig. 5C). For example, sample PD 3.4A (Fig. 5A) shows a distinct trend to a reversed southerly direction, but the inclination is zero for the end cluster, while sample PD 3.6A has an upward inclination from the beginning, but the declination along the great circle tends to a southerly direction (Fig. 5B). Combination of the six great circle planes for

Table 3
Summary of paleomagnetic results in (Lower) Pliocene marls covering the Calabro-Peloritan block

	Pol	D(tc)	I(tc)	k(tc)	Age	Locality
(Lower) Pliocene Trubi marls						
8 intervals	NR	14.4	42.4	173	Plio(L)	lower Singa (CSL)
4 sites	NR	13.3	34.8	80	Plio(L)	Careri (CAR)
4 sites	N	13.3	45.0	95	Plio(L)	Monte Límina (ML)
8 intervals	NR	3.9	45.2	59	Plio(L)	Capo Spartivento (CP)
7 sites	NR	9.0	49.4	272	Plio(L)	Paradísoni (PD)
4 sites	N	21.0	49.8	149	Plio(L)	Capo Vaticano (VAT)
3 sites	R	10.4	49.5	610	Plio(L)	Messina (MES)
3 sites	R	26.8	56.0	18	Plio	Villafranca(VIF)
4 localities	NR	11.3	41.9	180	Plio(L)	Ionian coast
4 localities	NR	16.4	51.4	173	Plio(L)	Tyrrhenian coast
8 localities	NR	13.7	46.7	108	Plio(L)	cover CaPe block
Results Aïfa et al. (1988)						
20 samples	R	92.7	59.8	20	Plio(L)	Capo dell'Armi
50 samples	R	18.9	51.5	7	Plio(L)	Quatro-Fave

Pol = polarity (N = normal, R = reversed, NR = both polarities); Plio = Pliocene, (L) = lower.

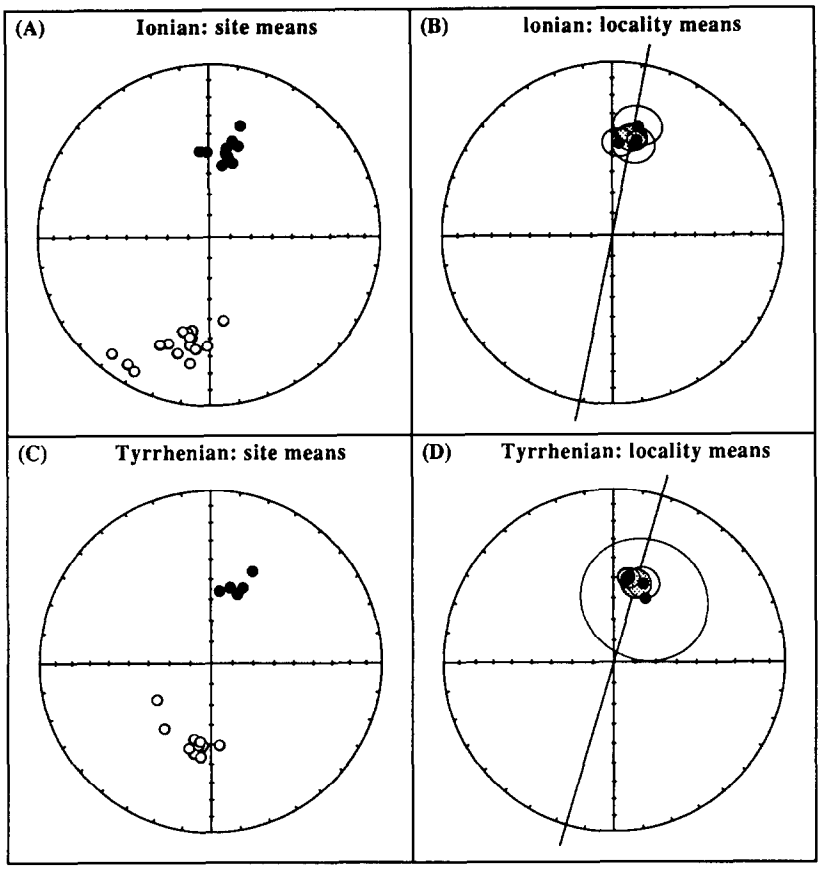
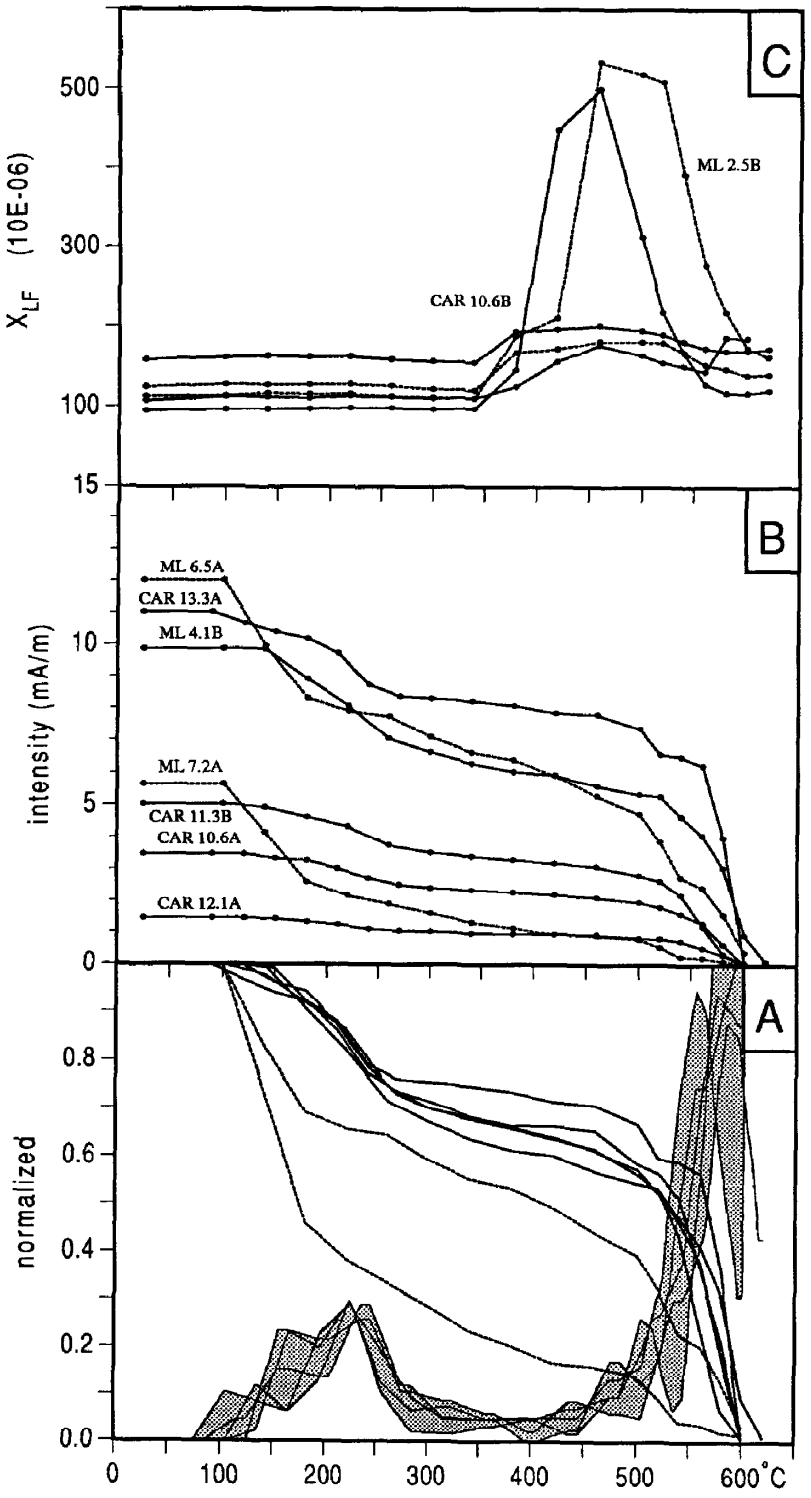


Fig. 6. Equal-area projection of site means of the HT component from the Lower Pliocene sediments along the Ionian and Tyrrhenian coast of both Calabria and Sicily (after bedding tilt correction). The site means are for all sites; the locality means were calculated using the sites of that locality. There is no significant rotation between the two groups of localities. Further see caption to Fig. 3.

this site gives a very well determined mean direction having a 7° clockwise rotation (Table 2; Fig. 5C). The other sites from the quarry PD[A] give similar results. Only site PD 5 has a normal polarity (Fig. 5D); its direction is anti-parallel to the calculated mean for the reversed polarity sites (Table 2).

The other quarry—PD[B]—containing sites PD 7–9 has initial NRM intensities up to 900 μA/m. Application of the great circle method shows that the mean directions are in good agreement with the results from the first quarry (Tables 2, 3). Calculation of a mean direction for the Lower Pliocene marls of the Paradísoni area us-

Fig. 7. Thermal NRM decay curves and behaviour of bulk susceptibility for samples from the Trubi formation (Spartivento basin, southern Calabria); the dashed lines represent samples having dirty HT components. Diagram (B) gives absolute intensities; diagram (A) shows normalised intensity curves. Shaded area is the area between the curves showing the general rate of decay (blocking-temperature spectrum), using a rectangular 40°C window. Diagram (C) displays the bulk susceptibility (χ) during thermal demagnetisation; there is a general increase in χ at 350°C.


C
B
A
X_{LF} (10E-06)
500
300
100
15
ML 2.5B
CAR 10.6B
intensity (mA/m)
10
5
0
ML 6.5A
CAR 13.3A
ML 4.1B
ML 7.2A
CAR 11.3B
CAR 10.6A
CAR 12.1A
normalized
0.8
0.6
0.4
0.2
0.0
0
100
200
300
400
500
600°C

ing the seven site means results in: $D_m = 9.0°$ and $I_m = 49.4°$ ($k = 272$, Table 2), implying a 9° clockwise rotation.

The Capo Vaticano section. For the Capo Vaticano section, initial NRM intensities are very low for all sampled sites (50–180 μA/m), except for site VAT 2 (7.6 mA/m). The other six sites could not be analysed using great circles, because the samples have a clear normal-polarity HT magnetisation and there is probably not a secondary magnetisation interfering with the HT component. Therefore, we used the stacking method (Scheepers and Zijderveld, 1992) to derive reliable directions per site. It appears that sites VAT 3–5 give satisfactory stacked diagrams, because of the relatively low α_{95} for the stacked sum vector. These three sites give clockwise rotation in accordance to the rotation observed for site VAT 2 (Table 2; Fig. 6). This supports the reliability of the low-intensity HT directions derived from the stacking method. Sites VAT 1 and VAT 6–7 were (even after stacking) too scattered and inconsistent to determine reliable HT directions. The mean direction of the four sites/levels is: $D_m = 21.0°$ and $I_m = 49.8°$ ($k = 149$) (Table 2); implying a 21° clockwise rotation since the Early Pliocene. The results of the two localities in Lower Pliocene marls along the Tyrrhenian coast of southern Calabria give clockwise rotations; averaging the eleven sites gives a 13° clockwise rotation (Table 2), which is nearly identical to the result from the Spartivento basin (12° clockwise).

The Messina and Villafranca areas (Sicily). The sites in marls on the Peloritan massif of northeastern Sicily have very low NRM intensities as well. There is only one site with high NRM intensities: site VIF 4, showing a 35° clockwise rotation (table 2; Fig. 5F). For the other sites the great circle method—sometimes in combination with direct observations—worked very well. The stacking method was performed for site VIF 5, because almost no secondary magnetisation is present. For the calculation of a mean direction for the Messina area, the outcrop containing sites MES 1–3 was considered as one site because of the insufficient data per site. The mean direction of the Messina area shows a consistent 10° clockwise rotation (Table 2). For the Villafranca area the two sites in Upper Pliocene marls (VIF 4–5) give clear clockwise rotations (Table 2), whereas the outcrop containing the Lower Pliocene sites VIF 2–3 shows a small counterclockwise deviation (table 2). Averaging the six sites of Pliocene age covering the Peloritan massif of Sicily implies an 18° clockwise rotation (Table 2): $D_m = 197.9°$ and $I_m = -53.0°$ ($k = 35$).

3.3. Magnetic carriers Trubi marls

The NRM decay curves (Fig. 7) clearly show that the HT component in the Trubi sediments is carried by a mineral with a maximum blocking temperature of 580–600°C. Samples without a secondary overprint, show nearly no decay in NRM between 300 and 450°C. The NRM decays significantly between 520 and 600°C, indicating that magnetite is the main magnetic mineral carrying the HT magnetisation. The decay curves show that part of the NRM is carried by a mineral with a maximum unblocking temperature of 250–270°C (Fig. 7A). Samples with a secondary overprint loose a large (30–50%) part of the magnetisation between 100 and 200°C (Fig. 7A). The bulk susceptibility (χ) during thermal demagnetisation (Fig. 7C) shows a very uniform behaviour up to 350°C; from this temperature (up to 450°C) χ increases considerably. This probably indicates the growth of magnetite grains (see also Van Hoof and Langereis, 1992). At higher temperatures, χ decreases to almost its initial value again (Fig. 7C).

For a number of samples, hysteresis measurements were performed to derive some supplementary rock magnetic properties. From these measurements, the coercive force (H_C) and saturation magnetisation (J_S), were determined using the method of Jackson et al. (1990). Hysteresis curves were determined by measuring the imbalance of a pair of coils, one of which contained the specimen, using an alternating field with a maximum of 250 mT. All curves show in high fields a linear magnetisation/field behaviour, which is caused by the paramagnetic contribution of the clay minerals. The slope of this linear increase at

high fields is used to calculate the high-field susceptibility (χ_{HF}) (Table 4; Fig. 8A). For the marls from the Trubi formation, χ_{HF} has a mean value of $(66 \pm 18) \times 10^{-6}$ SI units. The low-field susceptibility (χ_{LF}) was determined on a KLY-2 susceptibility bridge for all samples. The difference ("d") between the χ_{LF} and the χ_{HF} was calculated for all sites; it is a measure for the amount of remanence carrying minerals.

For the determination of the remanent coercivity (H_{CR}), the paramagnetic contribution was subtracted (Table 4); the samples having a $J_{RS} < 1$ A/m were not used to calculate the mean H_{CR} values because of their large inaccuracy. The reliable samples ($J_{RS} > 1$ A/m) have H_{CR} values between 39.3 and 52.1 mT, with a mean value of 46.2 mT (± 5.1 mT). The mean ratios $(H_{CR}/H_C) = 1.92$ (± 0.28) and $(J_{RS}/J_S) = 0.40$ (± 0.08) (Ta-

Table 4
Rock magnetic properties of some marly sites from this study deduced from hysteresis measurements

Specimen	H_{CR}	H_C	J_{RS}	J_S	*(A)*	*(B)*	χ_{LF}	χ_{HF}	d	*(C)*
$J_{RS} < 1$ A/m										
CAR 9.4A	127.6	21.5	0.19	1.05	5.92	0.18	78	70	8	41
CAR 12.2B	52.1	24.4	0.79	2.20	2.13	0.36	85	48	37	47
ML 7.4C	51.0	20.1	0.38	1.33	2.54	0.29	75	57	18	48
PD 1.5B	95.6	24.4	0.07	0.24	3.92	0.29	42	32	10	148
PD 2.1B	92.2	41.4	0.23	0.36	2.23	0.65	62	60	2	8
PD 3.2B	74.4	17.2	0.33	1.21	4.31	0.28	50	32	18	55
PD 4.1B	122.6	37.1	0.13	0.42	3.30	0.30	60	52	8	64
PD 5.2B	74.7	24.0	0.25	0.74	3.12	0.33	59	50	9	35
PD 6.5C	171.0	33.0	0.13	0.47	5.18	0.28	66	58	8	64
PD 7.4B	93.3	22.9	0.27	0.81	4.08	0.33	86	76	10	38
PD 8.1B	166.3	29.3	0.18	0.56	5.68	0.32	78	72	6	35
PD 9.6B	116.8	26.8	0.15	0.47	4.36	0.33	70	62	8	52
VAT 4.2B	70.5	16.0	0.46	1.94	4.41	0.24	70	46	24	52
VAT 6.4B	147.1	19.2	0.14	0.94	7.66	0.15	75	57	18	126
VIF 2.2B	169.2	49.7	0.16	0.38	3.41	0.43	68	52	16	102
VIF 3.5B	90.9	19.8	0.09	0.38	4.59	0.23	59	51	9	94
VIF 5.6B	171.2	24.8	0.20	0.81	6.90	0.24	97	82	15	75
$J_{RS} > 1$ A/m										
CAR 10.2B	51.0	26.7	1.05	2.68	1.91	0.39	163	62	101	96
CAR 11.6B	44.8	22.6	1.31	3.44	1.99	0.38	161	62	99	75
CAR 13.2B	52.1	26.7	1.39	3.44	1.95	0.40	213	84	129	93
ML 1.1B	47.6	29.8	1.49	2.98	1.60	0.50	131	86	45	30
ML 2.2B	45.5	28.6	1.52	2.96	1.59	0.51	119	76	43	28
ML 3.2C	49.4	28.1	1.43	3.04	1.76	0.47	139	82	57	40
ML4.6B	50.8	27.3	1.68	4.18	1.86	0.40	158	101	57	34
ML 5.2	42.9	21.5	1.07	2.95	2.00	0.36	106	64	43	40
ML 6.4C	36.0	21.7	1.99	4.34	1.66	0.46	148	69	79	40
VAT 2.1B	39.3	16.5	1.86	6.46	2.38	0.29	149	53	96	51
VIF 4.2B	49.0	20.4	1.67	5.86	2.40	0.28	186	106	80	48
Statistics										
Mean	46.2	24.5	1.50	3.85	1.92	0.40	152	77	75	52
S.d.	5.1	4.2	0.30	1.26	0.28	0.08	30	17	29	25
S.d. (%)	11	17	20	33	14	19	20	22	38	47

$H_{C(R)}$ = (remanent) coercivity force (mT); $J_{(R)S}$ = (remanent) saturation magnetisation; *(A)* = H_{CR} to H_C ratio, *(B)* = J_{RS} to J_S ratio, χ_{HF} = high-field susceptibility (SI units); χ_{LF} = low-field susceptibility (SI units); d is the difference between the low- and high-field susceptibility; (C) = d to J_{RS} ratio (in μm/A).

ble 4; Fig. 8B) show that the magnetite is of pseudo single-domain magnetite size (Day et al., 1977). The earlier studied Pleistocene clays of the Matera area in the Southern Apennines (Scheepers et al., 1993)—which have maximum blocking temperatures of 450–480°C—have distinct higher (J_{RS}/J_S) and lower (H_{CR}/H_C) ratios (Fig. 8B), indicating single-domain magnetite.

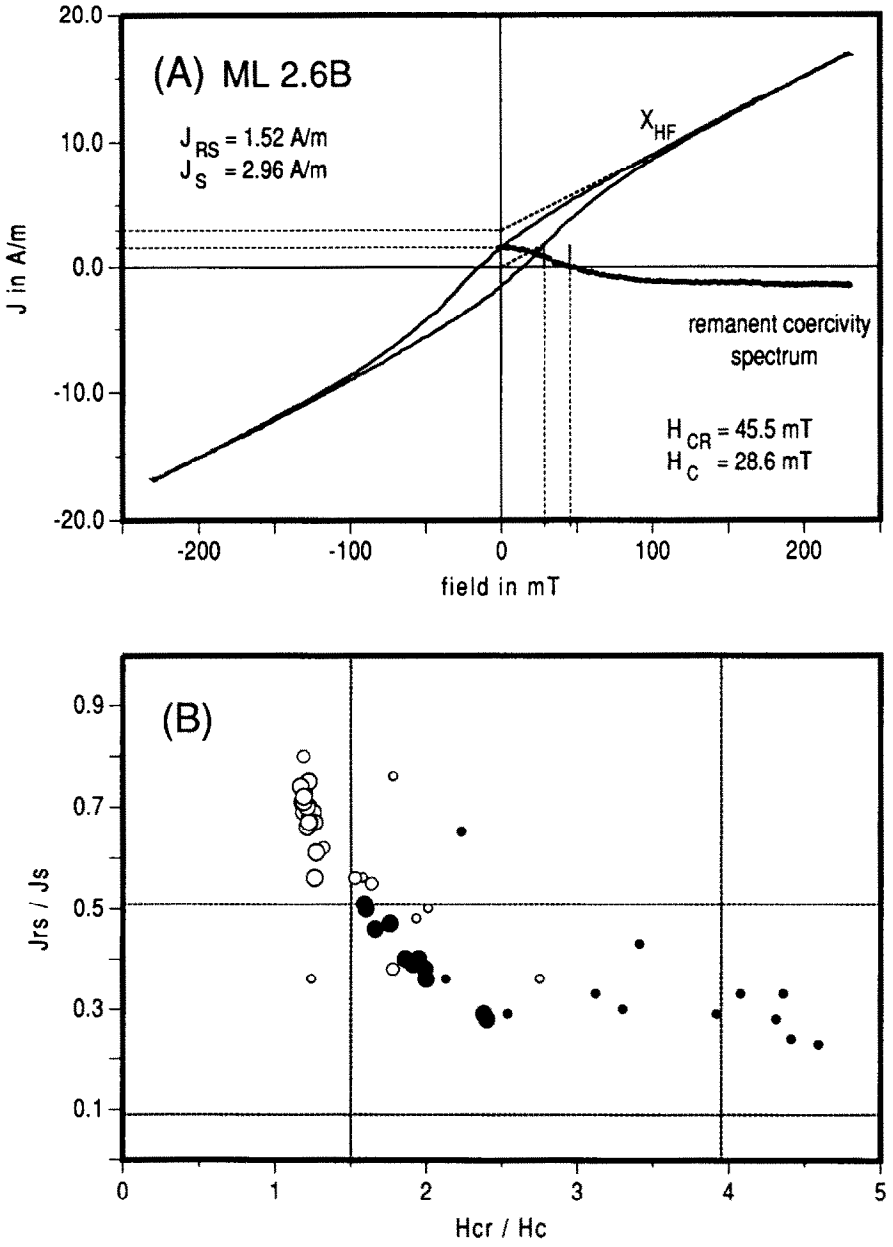


Fig. 8. (A) Magnetic hysteresis loop for clay sample ML 2.6B. Dotted line gives the remanent coercivity spectrum of the sample. The high-field linear magnetisation/field behaviour is caused by the paramagnetic contribution of the clay minerals. (B) the J_{RS}/J_S ratio versus the H_{CR}/H_C ratio for the Trubi marls (•) and the Pleistocene clays of the Matera area (○). Large (small) symbols represent samples with a large (small) J_{RS}.

4. The Upper Pliocene and Pleistocene clays

To study the tectonic rotation of younger sediments, Upper Pliocene and Pleistocene clays were sampled in four different areas covering and bordering the Calabro-Peloritan block. These areas are (Fig. 1): (a) the upper part of the Singa section in the Spartivento basin of southern Calabria; (b) one quarry near Croce Valanidi (southern Calabria near Reggio di Calabria); (c) two sections in the Catanzaro depression; and (d) two quarries east of Milazzo on Sicily just along the Tyrrhenian coast. Two of the already discussed sites from the Villafranca area are of Late Pliocene age; they show significant clockwise rotations.

The upper part of the Singa section from the Spartivento basin (southern Calabria) was studied earlier for magnetostratigraphy (Zijderveld et al., 1991). It is the same unit as described in section "the Ionian coast"; the upper part was sampled at 226 levels. The magnetostratigraphy shows that the age of the upper Singa section is between 2.5 and 1.9 Ma; the Gauss–Matuyama polarity transition occurs in the lower part of the section (Zijderveld et al., 1991). The bedding tilt is approximately 5° to the south-southeast.

At the extreme southwest of Calabria near

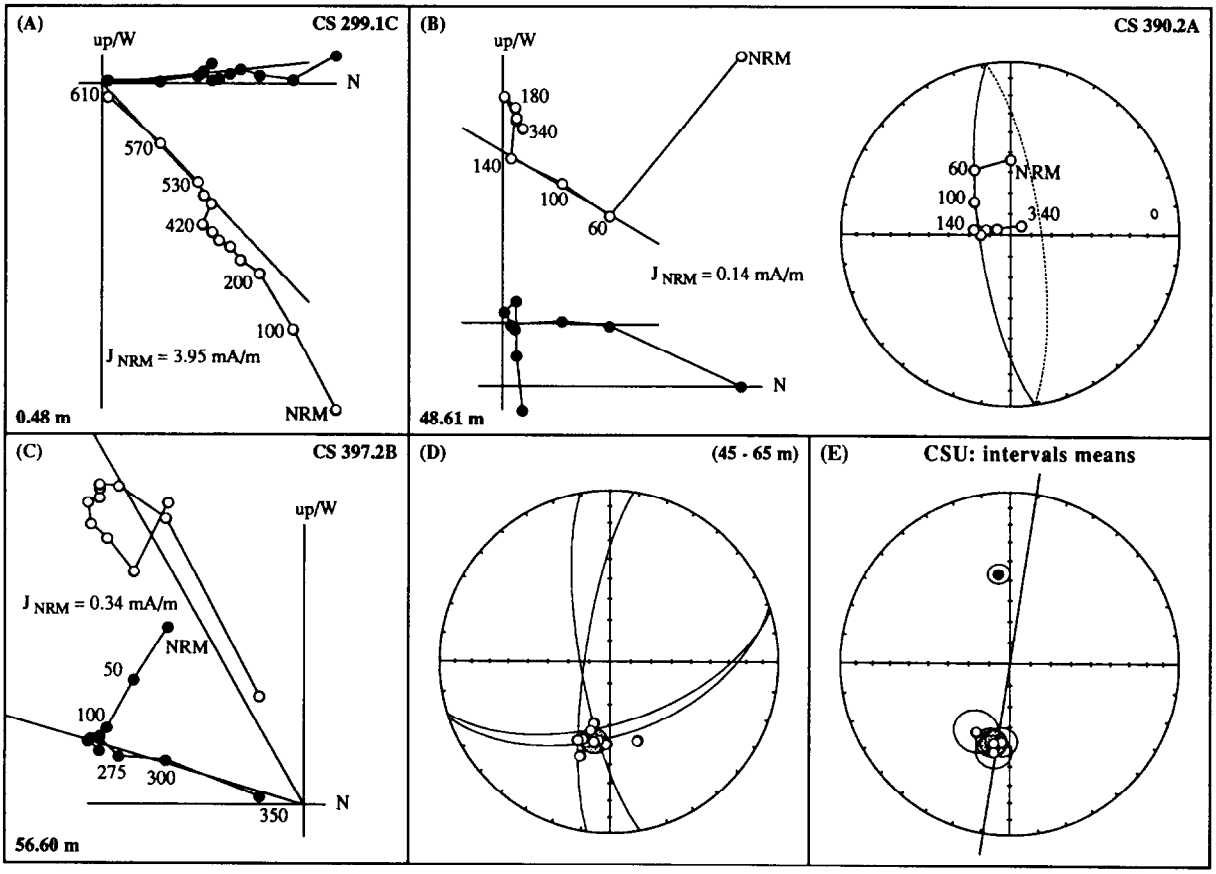


Fig. 9. Representative thermal demagnetisation diagrams and great circle planes of Upper Pliocene and Lower Pleistocene clays from the upper Singa section (southern Calabria). (E) represents the intervals means and the estimated clockwise deviation (8°) of the section. Symbols as in Figs. 2 and 4.

Reggio di Calabria (Fig. 1), we sampled ten sites in the middle Pleistocene clays of the Croce Valanidi quarry. This quarry was studied earlier by Aïfa et al. (1988) who suggested that the Jaramillo subchron is present in this quarry. According to our data the quarry is slightly younger, between the Brunhes–Matuyama and the top of the Jaramillo (see Appendix). Two other sections were sampled in the Catanzaro depression (middle Calabria; Fig. 1): Upper Pliocene clays near Caraffa di Catanzaro and Pliocene–Pleistocene silts near Santa Maria di Catanzaro. The silty sediments west of the village Santa Maria di Catanzaro are poorly exposed; outcrops are found along the road that goes from the village to the southwest. This section was earlier studied paleomagnetically by Watkins et al. (1974); according to these workers the Olduvai subchron (Gilsa event at that time) is recorded here. Sites SMC 3–4 are along the road in the upper part of the formation (levels 1–22 of Watkins et al., 1974); they were drilled in blue-coloured (clayey) silts. Sites SMC 1–2 are further to the southwest in a quarry some 40 m lower in the section; site SMC 5 is the lowest site in the section (10 m below SMC 1). The bedding plane has a tilt of 11° to the west for this quarry. The Caraffa di Catanzaro section is in Upper Pliocene clays east of the village; bedding tilt is approximately 5° to the southwest. The sampled interval is between 3.0 and 2.3 Ma (see Appendix); eight sites (CC 1–8) were drilled here. Along the Tyrrhenian coast of Sicily in the area east of Milazzo (Fig. 1) at the western part of the Calabro-Peloritan block, we sampled two quarries in very fresh blue-grey coloured clays of middle–Late Pleistocene age. Sites MIL 1–5 are in a quarry near Venético Marina, about 10 km east of Milazzo. Sites MIL 6–8 are in an outcrop south of Villafranca, about 7 km east of the quarry of sites MIL 1–5. The absolute age is between 0.6 and 0.4 Ma (see Appendix); there is no significant bedding tilt for both quarries.

Demagnetisation

For the younger Plio-Pleistocene clays there are two distinct types of demagnetisation behaviour: some sites have a HT component removed at 625°C (Fig. 10E–H), the other sites lose their magnetisation at 360–400°C (Fig. 10A–D, I).

The upper Singa section. The samples from the topmost part of this section have a demagnetisation behaviour which is completely different from the lower part; the uppermost (105–152 m) part of this section was not used in this study because the magnetostratigraphic analysis showed unreliable results (Zijderveld et al., 1991), because of intense weathering. The maximum blocking temperature of the samples from the Gauss Chron (lower part) is 580–600°C (Fig. 9A), which is identical to the results from the lower Singa section. The mean direction from this polarity zone has a small counterclockwise deviation (7°, Table 1). For the rest of the section, the magnetisation is removed at approximately 360°C (Fig. 9C), indicating different magnetic carriers of the NRM (probably iron sulphides). There are 36 samples which show clear reversed polarity magnetisations without dominant secondary components (see, for example, Fig. 9C); there is a linear decay to the origin from 100–150°C onwards; calculation of the mean direction gives a clockwise rotation (12°) for these samples (Table 1). Between the 25-m and 105-m level, the section is divided in four 20-m intervals; great circle planes were calculated for all the samples where a dominant secondary magnetisation made it impossible to calculate a reliable final component (see Fig. 9B). The combined analysis of these great circle planes and direct observations (McFadden and McElhinny, 1988) results in four mean directions (Fig. 9D; table 1). Calculation of an overall mean direction for the reversed polarity intervals gives a consistent clockwise deviation of 14° (Table 1; Fig. 9E); this is nearly identical to the mean declination of the lower Singa section. The normal polarity Gauss Chron interval has a counterclockwise deviation. The overall mean direction —which is used as an estimate for the tectonic rotation for the upper Singa section—is: D_m = 9.1° and I_m = 50.5° (k = 88), implying an 9° clockwise rotation (Table 1). This mean direction is not significantly different from the mean direction of the lower Singa, especially taking into account the consistent 14° clockwise rotation in

the Matuyama compared to the 14° clockwise rotation in the lower Singa section. It is (only) the upper part of the Gauss Chron which gives significantly deviating directions.

Croce Valanidi. According to Aïfa et al. (1988) the Jaramillo subchron is recorded in the Croce Valanidi clays. Our sites cover the whole quarry and initial NRM intensities show a large variety,

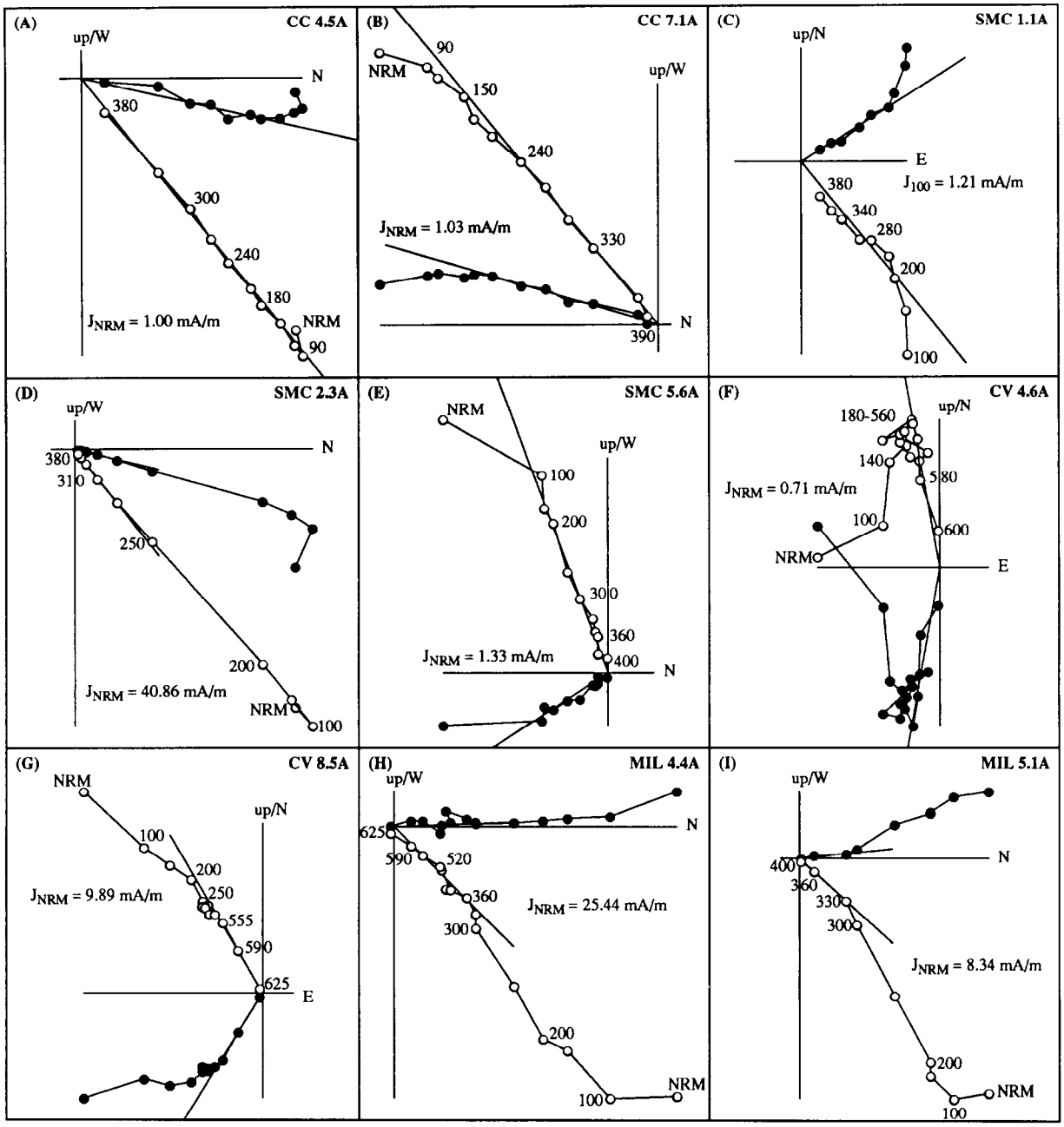


Fig. 10. Representative thermal demagnetisation diagrams of Upper Pliocene and Lower Pleistocene clays from southern Calabria. Symbols as in Fig. 2. *CC* = Caraffa di Catanzaro; *SMC* = Santa Maria di Catanzaro; *CV* = Croce Valanidi; *MIL* = Milazzo.

between 60 μA/m and 10.2 mA/m. Four out of ten sites give very good results (Fig. 10C, D), all having a reversed polarity for the HT component. The remaining six sites have (extremely) low remanences and were not used in this analysis; most sites have at least one sample which clearly tends to reversed polarities. From our analysis, it appears that the whole quarry has a reversed polarity. For the four reliable sites the magnetisation is removed above 600°C; the component removed between 500 and 600°C gives a clockwise deviation for all sites (Table 5). The mean direction of the four sites implies a 18.5° clockwise rotation for these middle Pleistocene clays.

Table 5
Paleomagnetic results from Upper Pliocene and Pleistocene clays covering the Calabro-Peloritan block using Fisher (1953) statistics

Site	Method	N	D(tc)	I(tc)	k	α95	Int
Croce Valanidi (Reggio di Calabria)							
CV 4	direct	7	197.6	−45.6	65	7.5	0.68
CV 7	direct	6	198.7	−39.6	322	3.7	3.35
CV 8	direct	6	208.6	−38.6	77	7.7	3.53
CV 10	direct	6	186.2	−52.9	161	5.3	3.23
Sites CV		4	198.5	−44.4	80		
Milazzo (Tyrrhenian coast, Sicily)							
MIL 1	direct	6	354.6	46.6	63	8.5	3.82
MIL 2	direct	5	16.1	63.9	298	4.4	2.37
MIL 3	direct	7	10.3	50.0	182	4.5	1.73
MIL 4	direct	7	350.8	49.2	146	5.0	8.46
MIL 5	direct	6	7.5	47.7	89	7.1	2.20
Quarry [A]		5	2.9	51.8	72	9.1	
MIL 6	direct	5	2.2	48.0	294	4.5	1.96
MIL 7	direct	6	355.6	49.0	434	3.2	7.20
MIL 8	direct	6	6.9	46.9	62	8.6	3.64
Quarry [B]		3	1.6	48.1	424	6.0	
Sites (MIL)		8	2.4	50.4	109	5.3	
Caraffa di Catanzaro (Catanzaro depression)							
CC 4	direct	8	8.5	50.6	48	8.1	0.41
CC 5	great circle	5	196.3	−49.3	7	31.1	0.04
CC 6	great circle	10	194.8	−36.8	22	10.6	0.20
CC 7	direct	7	191.8	−45.3	59	7.9	0.50
CC 8	great circle	6	186.6	−63.8	7	26.3	0.15
Sites (R)		2	193.4	−41.1	170	19.3	
Sites (NR)		3	11.9	44.3	123	11.2	
Santa Maria di Catanzaro (Catanzaro depression)							
SMC 1 [a]	direct	6	53.7	44.7	236	4.4	0.68
SMC 2	direct	6	11.1	50.0	1050	2.1	6.87
SMC 3	direct	6	13.3	50.7	625	2.7	0.38
SMC 4	direct	6	13.9	60.1	52	9.4	0.29
SMC 5 [a]	direct	6	140.8	−61.0	23	14.3	0.67
Sites (N)		3	12.7	53.6	202	8.7	

[a] Denotes (mean) direction not used to calculate overall mean direction for the locality.
Symbols are the same as in Tables 1 and 2.

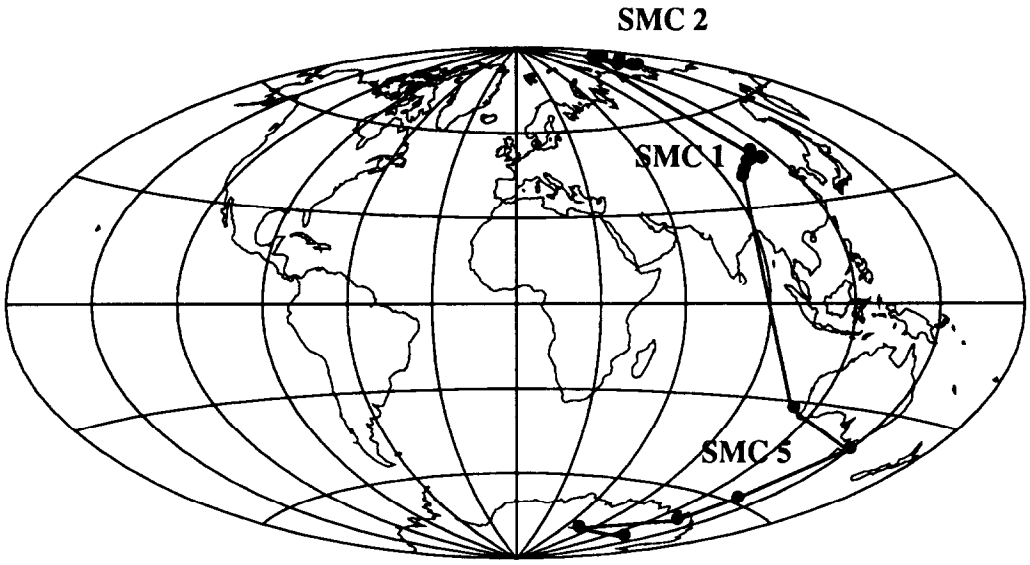


Fig. 11. Aitoff projection of virtual geomagnetic poles (VGP) from lower sites of Santa Maria di Catanzaro section (*SMC 5, 1* and *2*), most probably representing intermediate directions of the Matuyama–Brunhes transition.

Table 6
Summary of paleomagnetic results in Plio-Pleistocene sediments covering the Calabro-Peloritan block

	Pol	*D*(tc)	*I*(tc)	*k*(tc)	Age	Locality	Ref.
Plio-Pleistocene clays (Table 5)							
8 sites	N	2.4	50.4	109	Pleisto(M)	Milazzo	TS
4 sites	R	18.5	44.4	80	Pleisto(L)	Croce Valanidi	TS
5 intervals	NR	9.1	50.5	91	Plio(U)	Upper Singa	TS
3 sites	NR	11.9	44.3	123	Plio(U)	Caraffa di Catanzaro	TS
3 sites	N	12.7	53.6	202	Pleisto(L)	S. Maria di Catanzaro	TS
4 localities	NR	13.2	483	231	Pleisto(L)	Cover CaPe block	TS
(Lower) Pliocene Trubi marls (Table 3)							
8 localities	NR	14.0	46.8	108	Plio(L)	Cover CaPe block	TS
Results total Plio-Pleistocene cover							
12 localities	NR	13.7	47.3	140	Plio(L)–Pleisto(L)	cover CaPc block	
Other paleomagnetic results Calabria							
22 levels	N	17.4	53.9	48	Pleisto(l.)	S. MariadiCatanzaro	W.74
31 samples	N	12.0	44.4	28	Pleisto	Lazzaro-C	A.88
20 samples	R	92.7	59.8	20	Plio(L)	Capo dell'Armi	A.88
50 samples	R	18.9	51.5	7	Plio(L)	Quatro-Fave	A.88
Paleomagnetic results Southern Apennines							
3 localities	NR	336.6	57.2	254	Pleisto(L)	Matera area	S.93
Paleomagnetic results Caltanissetta basin (Sicily)							
2 intervals	NR	24.2	39.9	615	Plio(U)	Rossello section	SL.93
10 intervals	NR	34.2	46.4	1212	Plio(L)	Rossello section	SL.93

Pol = magnetic polarity (N = normal, R = reversed, NR = both polarities). Age: Plio = Pliocene; Pleisto = Pleistocene; (U) = Upper; (L) = Lower; (M) = middle. Ref.: TS = this study; W.74 = Watkins et al. (1974); A.88 = Aïfa et al. (1988); S.93 = Scheepers et al. (1993); SL.93 = Scheepers and Langereis (1993).

The Caraffa di Catanzaro section. For the Caraffa di Catanzaro section, initial NRM intensities are between 50 μA/m and 1.2 mA/m; only two sites give good results with the possibility of calculating the final magnetisation component directly (Fig. 10A, B). Site CC 4 in the middle part of the section has a normal polarity; site CC 7 in the upper part of the section has a reversed polarity. The remaining sites were analysed by using the great circle method, showing that all sites above CC 4 are clearly of reversed polarity. The calculated mean directions for these sites correspond very well with site CC 7. For these clays, the magnetisation is removed at 380–390°C, and NRM characteristics are the same as for the upper Singa section. The mean direction of the five sites gives a 12° clockwise rotation for the Upper Pliocene clays from Caraffa di Catanzaro: $D_m = 11.9°$ and $I_m = 44.3°$ (Table 5).

The Santa Maria di Catanzaro section. For the silty sites from the Santa Maria di Catanzaro section, initial NRM intensities are between 2.4 and 50.0 mA/m. The four upper sites (SMC 1–4) corresponding to the high-intensity levels 1–22 of Watkins et al. (1974) have a normal polarity end magnetisation; no secondary magnetisation is present here; all normal polarity sites have a clockwise rotation (Table 5). Site SMC 1 is the lowest and is strongly deviating from the others in its direction (Table 5; Fig. 10C), although it was taken in the same quarry as SMC 2 (Fig. 10D). The lower levels in the study of Watkins et al. (1974) give negative results because of the very low remanence intensities; nevertheless, they interpret all these levels as normal polarity. However, our lowest site SMC 5 is clearly of reversed polarity (Fig. 10E), although the mean declination is very unusual (141°) (Table 5; Fig. 10E). Together with the results of site SMC 1, we have probably recorded intermediate directions of the Matuyama–Brunhes transition (Fig. 11). Therefore, our estimate for the tectonic rotation is calculated by averaging only the upper three sites: $D_m = 12.7°$ and $I_m = 53.6°$ ($k = 202$, Table 5). The mean direction of the 22 reliable levels in the upper part of the section from the study of Watkins et al. (1974) corresponds very well to our mean direction (Table 6). The clockwise rotation is almost the same as that of the Caraffa di Catanzaro section (12°). The two sections in the Catanzaro depression thus clearly show clockwise rotations as well, confirming the hypothesis that the clockwise rotation is Pleistocene in age.

Milazzo. To give a more accurate age estimate for the tectonic rotation, middle–Late Pleistocene sediments were sampled on the Calabro-Peloritan block. The clays east of Milazzo, sites MIL 1–8, have high initial NRM intensities between 5 and 60 mA/m. One quarry has only sites with maximum blocking temperatures of 580–600°C (Fig. 10H), the other quarry shows maximum blocking temperatures of 350–400°C. The mean directions of both quarries correspond quite well in their declination: $D_m = 2.9°$ for quarry MIL[A] and $D_m = 1.6°$ for quarry MIL[B]. The overall mean direction of the eight sites is: $D_m = 2.4°$ and $I_m = 50.4°$ ($k = 109$, Table 5). Because the sites in Pliocene marls from the Peloritan massif have a mean direction showing an 18° clockwise rotation, the data from the Milazzo quarries are the first indication that the tectonic rotation had already taken place during the deposition of these sediments. We are aware, however, that more sites in these youngest sediments must be studied before we can reliably establish the rotation difference between the two time slices (middle and Late Pleistocene).

5. Discussion and conclusions

5.1. General results

Pliocene marls. For sites along the Ionian coast of southern Calabria it was possible to derive reliable HT magnetisation directions, indicating the tectonic rotation of the outcrop. However, the fact that a secondary component is present in both the lower part of the Monte Límina section and the Capo Spartivento section, and the HT directions are not anti-parallel, indicates that there is an overlap between the blocking-temperature spectra of the two NRM components (see also Van Velzen and Zijderveld, 1990). This over-

lap is still present in the HT component, determined between 510–540 and 600°C. Because the secondary component is of normal polarity (Brunhes), its influence will be much larger on reversed polarity HT directions, (certainly) if the section has a significant rotation. Therefore, it is probably not correct to simply average normal and reversed sites to give an estimate for the tectonic rotation. For Monte Límina, the best solution was to use the normal polarity sites for an estimate of the tectonic rotation. For Capo Spartivento, there is no difference in rotation of the different polarities; because this section has nearly no tectonic rotation the overlap has (only) influence on the inclination.

The four locations in Lower Pliocene marls of the Spartivento basin along the Ionian coast consistently reveal clockwise rotations (Table 3; Fig. 3). The mean direction giving equal weight to the locality mean directions is (Table 3): $D_m = 11.3°$ and $I_m = 41.9°$ ($k = 180$) (Fig. 6A, B). The mean inclination is consistently shallower than the geocentric axial dipole (GAD) inclination (57.5°) for the locality or than the present-day inclination (52°). This inclination error has been suggested to support the primary origin of the HT magnetisation in Greece (Laj et al., 1982). The clockwise rotation is much less than the 34° clockwise rotation derived from the same Early Pliocene formation in the more southwesterly Caltanissetta basin in Sicily (Scheepers and Langereis, 1993).

The four localities in (mostly Lower Pliocene) marls along the Tyrrhenian coast of both southern Calabria and Sicily have a mean direction: $D_m = 16.4°$ and $I_m = 51.4°$ ($k = 173$) (Fig. 6C, D). This is approximately the same as the mean direction for the Spartivento basin (Fig. 6B), although the mean inclination is steeper (51.4 vs. 39.8°), possibly implying less compaction for the Tyrrhenian sediments. Because declination values are practically the same for the two distributions (Ionian and Tyrrhenian), it appears that the whole Lower Pliocene cover of the Calabro-Peloritan block contains a uniform 14° clockwise rotation (Table 3). This is 20° less than the earlier established 34° clockwise rotation for the Lower Pliocene of the Caltanissetta basin on Sicily. The WNW-trending Taormina line (Amodio-Morelli et al., 1976; Moussat, 1983) in northeastern Sicily (Fig. 1) is proposed as the southern contact of the Calabro-Peloritan block, a contact with the rest of Sicily.

Younger Plio-Pleistocene clays. The four localities in Upper Pliocene and Lower Pleistocene sediments covering the Calabro-Peloritan block have a mean direction of: $D_m = 13.2°$ and $I_m = 48.3°$ ($k = 192$). This implies a rotation that is nearly identical to the rotation recorded in the (Lower) Pliocene marls (Table 6); this is supported by the observation that the mean declination of the Gilbert Chron of the lower Singa section is identical to the mean declination in the Matuyama Chron of the upper Singa section. So from the Singa section it is concluded that there are no phases of tectonic rotation for the Calabro-Peloritan block between 4.8 and 1.9 Ma. Averaging all twelve localities—both Pliocene and Lower Pleistocene—gives the best estimate for the total clockwise rotation of the Calabro-Peloritan block after the Early Pleistocene; this rotation is 13.7° clockwise (Table 6). According to the biostratigraphic age estimates of the sites from the Messina Strait (Croce Valanidi and Milazzo) and the Catanzaro depression (Santa Maria de Catanzaro), the rotation occurred between 0.7 and 0.2 Ma in the middle Pleistocene.

This middle Pleistocene consistent clockwise rotation of the Calabro-Peloritan block is the onset of major uplift of this block. Littoral deposits of Early–middle Pleistocene age crop out at an altitude of 1100 m in the Aspromonte of southern Calabria (Montenat et al., 1991). The same middle Pleistocene tectonic phase (major uplift and tectonic rotations) was earlier documented for the Matera area in the Southern Apennines (Scheepers et al., 1991, 1993; Sagnotti, 1992). The rotation is 23° counterclockwise in this (northern) part of the Tyrrhenian arc; this means a differential rotation of 37° between the two blocks for the middle Pleistocene rotation phase. The major tectonic lineament separating the Calabro-Peloritan block from the Southern Apennines is the Pollino Line, which is considered as a left lateral shear zone (Knott and Turco, 1991). It is further evident from this study and the studies

in the Southern Apennines that tectonic rotations in the Tyrrhenian arc system have occurred after the major phases of extension in the Tyrrhenian basin: the major phase of active spreading in the southeastern part of the Tyrrhenian basin was in the Early Pleistocene between 2.0 and 1.6 Ma (Sartori, 1990; Savelli and Schreider, 1991). Thus, there is clearly no simultaneous occurrence of extension/spreading in the Tyrrhenian basin and tectonic rotations in the surrounding arc.

Our hypothesis—for the time being—is that the whole area between the two major tectonic lineaments (Pollino Line to the north and Taormina Line to the south) behaved as one single block and performed a clockwise rotation of approximately 15° in the middle Pleistocene. More sites in Lower to middle Pleistocene sediments in northern Calabria are being studied to test this hypothesis.

5.2. Conclusions

From the paleomagnetic data it appears that the Calabro-Peloritan block—which forms the central part of the Tyrrhenian arc—underwent a consistent 15° clockwise rotation since the Early Pliocene. The rotation is still present in Upper Pliocene and Lower Pleistocene clays covering this block, but the Milazzo clays (Peloritan massif) from the youngest part of the middle Pleistocene do not show this clockwise rotation. Our conclusion is that the whole southern part of the Tyrrhenian Arc underwent a uniform 15° clockwise rotation in the middle Pleistocene. Extension/spreading in the Tyrrhenian basin and tectonic rotations in the Tyrrhenian arc were not simultaneous. Tectonic rotations occurred at the onset of the major uplift of the arc, approximately 1 Myr after the major phase of spreading in the southeastern part of the Tyrrhenian basin.

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Appendix

On the basis of the foraminiferal biostratigraphy, and the polarity and stratigraphic succession of the sites (Tables A1,

Table A1
Ages of biostratigraphic datum planes in the Mediterranean used in this study to determine the ages of paleomagnetic sites (based on Hilgen, 1991b; Zijderveld et al., 1991; Castradori, 1993)

Species	Event	Age (Ma)
Formaminiferal species		
Globorotalia truncatulinoides	re-appearance	1.05
Hyalinea balthica	first occurrence	1.45
sinistral neogloboquadrinids	first common occurrence	1.68
Globorotalia inflata	re-appearance	1.93
Globorotalia truncatulinoides	first occurrence (intlux)	1.95
Globorotalia inflata	first occurrence (intlux)	2.02
Neogloboquadrina atlantica	last occurrence	2.31
Globorotalia bononiensis	last occurrence	2.31
Neogloboquadrina atlantica	first occurrence	2.55
Globoquadrina altispira	last occurrence	3.04
sphaeroidinellopsis seminulina	last occurrence	3.07
Globorotalia margaritae	last occurrence	3.56
Globorotalia margaritae	last common occurrence	3.71
Globorotalia puncticulata	first occurrence	4.13
Globorotalia margaritae	first common occurrence	4.64
Sphaeroidinellopsis spp.	top acme	4.74
Sphaeroidinellopsis spp.	base acme	4.83
Nannofossil species		
P. lacunosa	last occurrence	0.46
Gephyrocapsa sp. 3	last occurrence	0.55
Gephyrocapsa sp. 3	first occurrence	0.90
Gephyrocapsa oceanica	re-appearance	0.90
Gephyrocapsa oceanica	first occurrence	1.60

Ages of datum planes are all based on the geomagnetic polarity time scale of Berggren et al. (1985) in anticipation of a more general acceptance of the astronomically calibrated time scale of Shackleton et al. (1990) and Hilgen (1991a,b). For this reason we recalculated the ages of the calcareous nannofossil datum planes presented in Castradori (1993).

Table A2
Polarity, planktonic foraminiferal biostratigraphy and ages of Pliocene sites studied. Ages are based on a combination of the polarity, biostratigraphy and stratigraphic order of sites

Site	Strike	Dip	Pol	1	2	3	4	5	6	7	8	Age (Ma)
CC 8	144	7	R					x			00:10	2.47–2.31
CC 7	144	7	R				x	x			01:09	2.47–2.31
CC 6	144	7	R		rew		x	x		x	01:09	2.47–2.31
CC 5	144	7	R		rew		x	x			00:10	3.07–2.31
CC 4	144	7	N				x				00:10	3.07–2.47
CC 3	144	7	N(?)					x			00:10	3.07–2.47
CC 2	144	7	N(?)				x				00:10	3.07–2.47
CC I	144	7	N(?)		rew	rew		x			00:10	3.07–2.47
ML 5	116	14	N								00:10	4.74–4.57
ML 4	116	14	N	r							00:10	4.74–4.57
ML 3	116	14	N	c							00:10	4.77–4.74
ML 2	116	14	R	c							00:10	4.83–4.77
ML 1	116	14	R	c							01:09	4.83–4.77
ML 7	116	14	R	c							03:10	4.83–4.77
ML 6	116	14	NR	r							04:10	4.86–4.83
CAR 1	74	11	R		c	x					10:20	4.10–3.71
CAR 2	74	11	?		c	x					01:09	4.10–3.71
CAR 3	74	11	R		c	x					00:10	4.10–3.71
CAR 4	74	11	R		c	x					01:09	4.10–3.71
CAR 5	74	11	?		c	x					00:10	4.13–3.71
CAR 6	74	11	?		c						00:10	4.40–4.13
CAR 7	74	11	?		c						01:09	4.40–4.13
CAR 8	74	11	R		c						01:09	4.40–4.24
CAR 9	74	11	R		c						00:10	4.40–4.24
CAR 10	74	11	R		c						00:10	4.40–4.24
CAR 11	74	11	N		c						00:10	4.47–4.40
CAR 12	76	9	R		c						00:10	4.57–4.47
CAR 13	96	9	R		c						00:10	4.57–4.47
PD 9	262	11	R	r							00:10	4.57–4.24
PD 8	262	11	R								00:10	4.57–4.24
PD 7	262	11	N								00:10	4.74–4.40
PD 6	325	3	(?)								00:10	4.74–4.40
PD 5	325	3	N	r					x		00:10	4.74–4.40
PD 4	325	3	R	c							00:10	4.83–4.77
PD 3	325	3	R	c							09:01	4.83–4.77
PD 2	325	3	R	c							00:10	4.83–4.77
PD 1	325	3	R	c							00:10	4.83–4.77
VAT 7	85	2	R(?)		c						00:10	4.57–4.13
VAT 6	85	2	R(?)	r	r						00:10	4.57–4.13
VAT 5	85	2	N		c				x		00:10	4.64–4.13
VAT 4	85	2	N		c				x		01:09	4.64–4.13
VAT 3	85	2	N		r						00:10	4.64–4.13
VAT 2	85	2	N		c						00:10	4.64–4.13
VAT 1	85	2	(?)		c						00:10	4.64–4.13
VIF 3	284	9	R		c						00:10	4.57–4.24
VIF 2	284	9	R		c						00:10	4.57–4.24
VIF 4	179	4	R		rew		x	x			00:10	3.18–2.31
VIF 5	186	6	R				x	x		x	00:10	2.47–2.31

Table A2 (continued)

Site	Strike	Dip	Pol	1	2	3	4	5	6	7	8	Age (Ma)
MES 1	250	12	R	c							00:10	4.83–4.77
MES 2	250	12	R	r	r	x					00:10	4.10–3.56
MES 3	250	12	R	c							10:15	4.83–4.77
MES 5	5	12	R	c							03:09	4.83–4.77
MES 4	5	12	R	c	r						17:10	4.83–4.77

1 = *Sphaeroidinellopsis* spp.; 2 = *Globorotalia margaritae*; 3 = *G. puncticulata*; 4 = *G. crassaformis*; 5 = *G. bononiensis*; 6 = *G. altispira*; 7 = *Neogloboquadrina atlantica*; 8 = neogloboquadrinids (sinistral:dextral).
Sites from a single locality are given in stratigraphic order if no extra space is left open between the sites. Ages of biostratigraphic datum planes used to calculate the ages of sites are given in Table A1. rew = reworked; c = common; r = rare.

A2, A3), very precise ages could be assigned to all Pliocene sites, but only approximate ages to the Pleistocene sites. Table A3 shows that the ages of the Pleistocene MIL, CV and SMC sites do not differ from each other. This problem of age differentiation mainly results from the lack of resolution of the (integrated benthonic and planktonic) foraminiferal biostratigraphy in this time interval. Moreover, the intermittent and rare occurrence of *G. truncatulinoides* after its re-appearance dated around 1.05 Ma in the Mediterranean precludes any attempt to try to locate this re-appearance level with certainty in sections CV and SMC. For SMC, it is well known from the literature that *G. truncatulinoides* enters the

Table A3
Polarity, planktonic foraminiferal and nannofossil biostratigraphy and ages of Pleistocene sites studied

Site	Strike	Dip	Pol	1	2	3	4	Age 1 (Ma)	5	6	7	Age 2 (Ma)
CV 6	194	7	R	x			00:10	1.45–0.73	x	x	3.0	0.92–0.73
CV 5	194	7	R	x			22:07	1.45–0.73			0.4	0.92–0.73
CV 4	194	7	R	x			00:10	1.45–0.73	x	x	1.0	0.92–0.73
CV 3	194	7	R	x		x	16:11	1.45–0.73			0.6	0.92–0.73
CV 2	194	7	R(?)	x		x	00:10	1.45–0.73	x	x	0.6	0.92–0.73
CV 1	194	7	R(?)	x		x	01:09	1.45–0.73			2.6	0.92–0.73
CV 9	194	7	(?)	x	1	x	00:10	1.05–0.73		x	1.0	0.92–0.73
CV 8	194	7	R	x			00:10	1.45–0.73	x	x	1.0	0.92–0.73
CV 7	194	7	R	x			01:09	1.45–0.73	x	x	1.2	0.92–0.73
CV 10	194	7	R	x		x	08:14	1.45–0.73		x	0.0	0.92–0.73
SMC 4	0	0	N	x		x	00:10	< 0.98		x	6.4	0.73
SMC 3	0	0	N	x			02:08	< 0.98		x	6.0	0.73
SMC 2	199	11	N	x	x	x	00:10	< 0.98		x	3.2	0.73
SMC 1	199	11	N	x	x	x	02:09	< 0.98		x	1.2	0.73
SMC 5	0	0	R	x		x	00:10	1.45–0.73	1	x	0.4	0.73
MIL 8	0	0	N	x			02:10	< 0.98		x	rew	0.46–0.20
MIL 7	0	0	N	x			07:16	< 0.98		x	rew	0.46–0.20
MIL 6	0	0	N	x	1		00:10	< 0.98		x	rew	0.46–0.20
MIL 3	0	0	N	x		x	01:09	< 1.10		x	rew	0.46–0.20
MIL 2	0	0	N	x			01:09	< 1.10		x	rew	0.46–0.20
MIL 1	0	0	N	x		x	01:09	< 1.10		x	rew	0.46–0.20
MIL 5	0	0	N	x			00:10	< 1.88		x	rew	0.46–0.20
MIL 4	0	0	N	x			00:10	< 1.88		x	rew	0.46–0.20

1 = *Globorotalia inflata*; 2 = G. *truncatulinoides*; 3 = *Hyalinea balthica*; 4 = neogloboquadrinids (sinistral:dextral); 5 = *Gephyrocapsa* sp. 3; 6 = *G. oceanica*; 7 = *P. lacunosa*; rew = reworked.
Ages are based on a combination of the polarity, biostratigraphy and stratigraphic order of sites. Sites from a single locality are given in stratigraphic order if no extra space is left open between the sites. Ages of biostratigraphic datum planes used to calculate the ages of sites are given in table A1. For *P. lacunosa*, the percentage is based on a counting of 500 individual coccoliths. The presence of *Gephyrocapsa* sp. 3 implies that the percentage of *Gephyrocapsa* sp.3 of all medium sized gephyrocapsids is > 1%.

record stratigraphically well below the "G" marker bed (Gignoux, 1913), whereas all our sites were taken above this bed. But the MIL sites are the only sites in the present study which lack the usually very consistent clockwise rotation of 15° and it is therefore crucial to determine temporal relations between the Pleistocene sites as accurately as possible. For this purpose, we decided to employ also calcareous nannofossils, because of the higher resolution of the nannofossil biostratigraphy in this particular interval. The presence/absence of the relevant marker species has been indicated in Table A3.

The overall reversed polarity in combination with the presence of *Gephyrocapsa* sp. 3 in a number of CV sites provides clear evidence that all CV sites belong to the reversed interval between the Brunhes–Matuyama and the top of the Jaramillo. Likewise, the overall normal polarity of the MIL sites in combination with the absence of Pleistocene *P. lacunosa* and sometimes ubiquitous *Gephyrocapsa oceanica* (*Gephyrocapsa* > 4 μm of Rio et al., 1990) indicates that these sites belong to the Brunhes. The absence of distinct *Gephyrocapsa* sp. 3 suggest that all MIL sites postdate the LAD of *Gephyrocapsa* sp. 3 dated at 0.55 Ma (Castradori, 1993; see also Table A1). The additional absence of Pleistocene *P. lacunosa* (the rare specimens present show the morphology of Pliocene *P. lacunosa* and are probably reworked) further suggest that the MIL sites postdate the LAD of *P. lacunosa* dated at 0.46 Ma as well. Finally, the sites predate the FO of *Emiliani huxleyi* because no distinct specimens of this taxon were encountered with the use of optical light microscopy. This observation provides a minimum age of approximately 0.2 Ma for these sites.

The ubiquitous presence of *G. oceanica* in combination with the presence of the planktonic foraminifer *G. truncatulinoides* and a single specimen of *Gephyrocapsa* sp. 3 (in SMC 5) indicates that the observed polarity reversal from R to N between SMC 5 and 1 most probably represents the Brunhes/Matuyama boundary. The near absence of *Gephyrocapsa* sp. 3 can be explained by the fact that this taxon is rarely encountered in the basal part of the Brunhes (see Castradori, 1993).

It is evident from the biostratigraphic dating that the MIL sites lacking the clockwise rotation are the youngest sites studied. If this rotation affected the Calabrian structural block as a whole, then the rotation must be younger than 0.73 Ma (i.e. the age of the SMC sites) but older than 0.2 Ma (the minimum age of the MIL sites).

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